Functional Specification

AI Customer Support Platform

Functional requirements for an AI-native customer support platform (text and voice conversations)

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Role: Product Owner and development team

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Status: Draft for validation

This document describes *what* the system must do (needs, functions, business rules, constraints). Design and technology choices are covered in the architecture documentation and the technology report.

Revision history and approval

Revision history

Version	Date	Description
0.1	—	Document outline and scope.
1.0	October 23, 2026	First complete version: functional requirements per epic, business rules, non-functional requirements, planning and acceptance.

Approval

Role	Name	Date	Signature
Author / Product Owner	Amine Essid		
Customer-support representative	[to be filled]		
Security and privacy lead	[to be filled]		
Supervisor / client	[to be filled]		

Status of the content

The backlog, Sprint order and priorities shown here are a *planning hypothesis*, not a commitment. Team size, capacity and calendar are not yet known. Numeric targets (latency, availability, volumes) are deliberately left as **TBD** until they are measured or agreed with stakeholders; this document does not invent them.

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Chapter 1

Introduction

1.1 Purpose of this document

This functional specification formalizes the functional needs of the **AI Customer Support Platform**: an AI-native customer support platform that conducts complete conversations over text and voice, answers from the organization's approved knowledge, performs explicitly authorized business actions, and hands the conversation to a human agent without losing context.

It is the shared reference between the Product Owner, the development team, customer-support representatives and security owners. It is the basis for acceptance (Chapter 11) and feeds the Scrum Product Backlog.

1.2 Scope of the document

The document covers:

- the context, objectives and scope of the product;
- the actors and their use cases;
- detailed functional requirements for the 12 epics and 20 logical modules;
- cross-cutting business rules and the high-level data model;
- user interfaces and interfaces with external systems;
- non-functional requirements (security, privacy, reliability, performance, observability, etc.);
- release planning, risks, constraints and the acceptance strategy.

Detailed architecture, technology choices and UML diagrams are described in the other project deliverables (see Section 1.5).

1.3 Intended audience

Reader	Expected use
Product Owner	Validate the need, order the backlog, arbitrate priorities.
Developers	Understand expected behaviours and the rules to implement.
Support staff / agents	Check that handling and handoff journeys fit their work.
Security and privacy	Validate tenant isolation, access management and retention.
Supervisor / client	Validate scope, deliverables and acceptance criteria.

1.4 Writing conventions

1.4.1 Requirement levels

- **MUST**: mandatory for correctness or security.
- **SHOULD**: recommended unless a documented reason exists.
- **MAY**: optional.

1.4.2 MoSCoW priorities

Each requirement carries a MoSCoW priority.

Code	Meaning	Interpretation
M	Must have	Essential for the targeted release.
S	Should have	Important, workable around in the short term.
C	Could have	Desirable if capacity allows.
W	Won't have (now)	Deferred beyond the stated release, already identified.

1.4.3 Releases

R1: text MVP; **R2**: voice MVP; **R3**: operational expansion (see Chapter 10).

1.4.4 Identifiers

Prefix	Meaning
`FR-xxx-nn`	Functional requirement (`ACC` access, `CNV` conversation, `WEB` web chat, `KB` knowledge, `IA` orchestration, `OUT` tools, `HUM` human hand-off, `TKT` tickets, `VOX` voice, `WKF` workflows, `INT` integrations, `OBS` observability).
`RG-xxx-nn`	Cross-cutting business rule.
`NFR-xxx-nn`	Non-functional requirement.
`US-nnn`	User story from the Product Backlog.
`Mnn`	Logical module (M01 to M20).
`Enn`	Epic (E01 to E12).

1.5 Reference documents

- Product vision and scope (`01-product-vision`).
- Module catalog (`02`), architecture (`03`), module communication (`04`), domain model (`05`).
- AI conversation engine (`06`), voice and multimodal (`07`), integrations and tools (`08`).
- Security, privacy and reliability (`09`), observability and evaluation (`10`).
- Technology stack (`11`–`13`) and Scrum artifacts (`scrum/`).
- Project video with voice-over (Appendix D), a 1 min 31 s introduction to the platform.

Diagrams in this document come from the design documentation.

1.6 Minimal glossary

A full glossary is given in Appendix A. The essential terms are:

Tenant a customer organization of the platform; the data-isolation boundary.

Conversation the durable record of an interaction; channel sessions are only the delivery mechanism.

Turn the processing of one customer message by the AI engine.

RAG retrieval-augmented generation from approved knowledge.

Tool a business operation the AI may *request* and that application code authorizes and executes.

Handoff transfer of the conversation from the AI to a human agent together with its context.

Chapter 2

Project overview

2.1 Context and problem statement

Customer-support teams face growing volumes of repetitive requests, expectations of instant answers and a multiplication of channels (website, voice, e-mail, messaging). Classic chatbots handle open questions poorly, do not always rely on trustworthy sources, and cannot act safely on business systems. A fully human service, on the other hand, is costly and slow for simple requests.

The project targets a platform that:

- conducts **complete, context-aware conversations**, first by text then by voice;
- answers from **approved knowledge** and acknowledges uncertainty;
- executes only **authorized, audited and controlled actions**, enforced by application code;
- guarantees a **human escape hatch**: the customer can always reach a person;
- strictly isolates the data of each organization.

2.2 Vision and product principles

Vision

Build an AI-native customer support platform that can conduct complete, context-aware customer conversations over text and voice, answer using trusted business knowledge, perform explicitly authorized actions, and transfer to human agents when appropriate.

Eight principles shape the requirements:

1. **Conversation-first**: a conversation is the durable interaction record; channel sessions are delivery mechanisms.
2. **Modality-independent intelligence**: text and voice share the same orchestration and authorization model.
3. **Grounded answers**: use approved knowledge and business data, and be transparent about uncertainty.
4. **Safe action execution**: models *propose*; application services *authorize and execute*.
5. **Human escape hatch**: customers can request a human, and policy can escalate automatically.
6. **Tenant isolation by default**: data, search, tools, events and analytics.
7. **Observable and testable**: every important AI decision, tool execution, handoff and failure is measurable.
8. **Provider portability**: model, speech, channel and integration providers are accessed through adapters.

2.3 Objectives

2.3.1 Product outcomes

- Reduce time to a correct answer.
- Resolve repetitive requests without removing access to human support.

- Preserve context across text, voice and channel changes where identity can be safely linked.
- Make AI actions auditable and permission-controlled.
- Let organizations configure knowledge, tools, policies and escalation behaviour.
- Measure *resolution quality*, not only response speed.

2.3.2 First milestone objective (MVP)

A website assistant can, end to end:

1. accept a customer text message;
2. maintain a conversation;
3. retrieve relevant organization knowledge;
4. invoke one safe, read-only business tool;
5. answer with source-grounded information;
6. escalate to a human while preserving context;
7. emit audit and observability events.

Voice *reuses* the same conversation engine and tool authorization model (release R2).

2.3.3 Success measures

Table 2.1: Product success measures

Measure	Definition in principle	Target
Correct answer rate	Share of answers judged correct on a curated evaluation set.	TBD
Groundedness rate	Share of claims supported by a source.	TBD
Task completion rate	Share of requests carried to completion (including via a tool).	TBD
Human escalation rate	Handoffs relative to volume, segmented by reason.	TBD
Median and p95 latency	Response time, for text then for voice.	TBD
Tool success rate	Successful calls over authorized calls.	TBD
Customer satisfaction	Customer feedback collected after a conversation.	TBD
Cost per resolved conversation	Provider and operating cost.	TBD
Availability and delivery	Service availability and message-delivery success.	TBD

Targets are to be set with stakeholders after a first baseline measurement.

2.4 Scope

2.4.1 Included in the MVP (R1)

- Organization, user and access foundation.
- Website text chat.
- Conversation lifecycle and message persistence.

- Basic customer identity.
- Knowledge source ingestion and retrieval (RAG).
- AI response orchestration.
- One read-only business tool (order lookup).
- Human inbox and handoff.
- Basic ticket creation.
- Audit, logs, metrics, traces and usage tracking.

2.4.2 Later phases

- **R2**: real-time browser voice, STT/TTS or speech-to-speech depending on the provider, transcript synchronization.
- **R3**: e-mail and external messaging channels, workflow builder, advanced routing and SLA, more connectors, write-capable tools, advanced analytics, evaluation, billing and enterprise governance.

2.4.3 Explicit non-goals for the first milestone

- Replacing all existing CRM and helpdesk systems.
- Arbitrary autonomous changes to customer accounts.
- Unreviewed financial transactions.
- Supporting every channel and provider at launch.
- Training a foundation model.
- Splitting every module into a microservice.

2.5 Stakeholders

Table 2.2: Stakeholders

Stakeholder	Main expectations
End customers	Fast and correct answers, access to a human, protected data.
Support agents	Conversation takeover with context, a clear work queue, efficient tools.
Support leads	Control of queues, policies, quality and SLAs.
Organization administrators	Configuration of channels, knowledge, AI agents, integrations and access.
Platform operators	Reliability, cost, abuse control and system health.
Security and privacy	Tenant isolation, data minimization, auditability.
Business-system owners	Controlled, limited access to their APIs.

2.6 General assumptions and constraints

2.6.1 Assumptions

- A small team initially builds a **modular monolith**; services are extracted only when scaling, fault isolation or deployment cadence justify it.
- Product design, infrastructure, security and automated tests are part of delivery, not later phases.
- Story points are team-specific estimates, not time conversions.

- Sprints are assumed to last two weeks; the sequence is provisional.

2.6.2 Constraints

- AI model output is **untrusted input**: authorization and business invariants are enforced by application code.
- The core domain depends on no specific model or messaging provider; provider details live in adapters.
- No raw audio or sensitive customer data is stored without a clear product need, consent or legal basis, access policy and retention policy.
- Every request or job carries a trace identifier and tenant context.

Chapter 3

Actors and use cases

3.1 Human actors

Table 3.1: Human actors

Actor	Role and goals
Customer	Asks a question or requests an action through a supported channel (web chat, then voice); wants a correct answer or a human.
Support agent	Reviews, takes over and resolves conversations or tickets; receives the handoff context.
Support lead	Manages queues, policies, reporting and quality; filters and follows tickets.
Organization administrator	Configures channels, knowledge, AI agents, integrations and access; invites users and assigns roles.
Platform operator	Monitors reliability, cost, abuse and platform health.

3.2 System actors

Table 3.2: External systems and technical actors

System	Interaction
Model provider (LLM)	Generation, classification, embeddings, through an adapter.
Speech provider (STT/TTS or speech-to-speech)	Transcription and speech synthesis (R2).
Channel providers	Webhooks and outbound delivery (e-mail, messaging; R3).
Business systems (commerce, CRM, etc.)	Tool calls through connectors, with limited permissions.
Secret manager	Reference and rotation of credentials.
Telemetry stack	Logs, metrics, traces, alerts.

3.3 Use cases

The following figures show the use cases per actor family. Each use case is detailed in Chapter 5 as requirements and in Appendix B as scenarios.

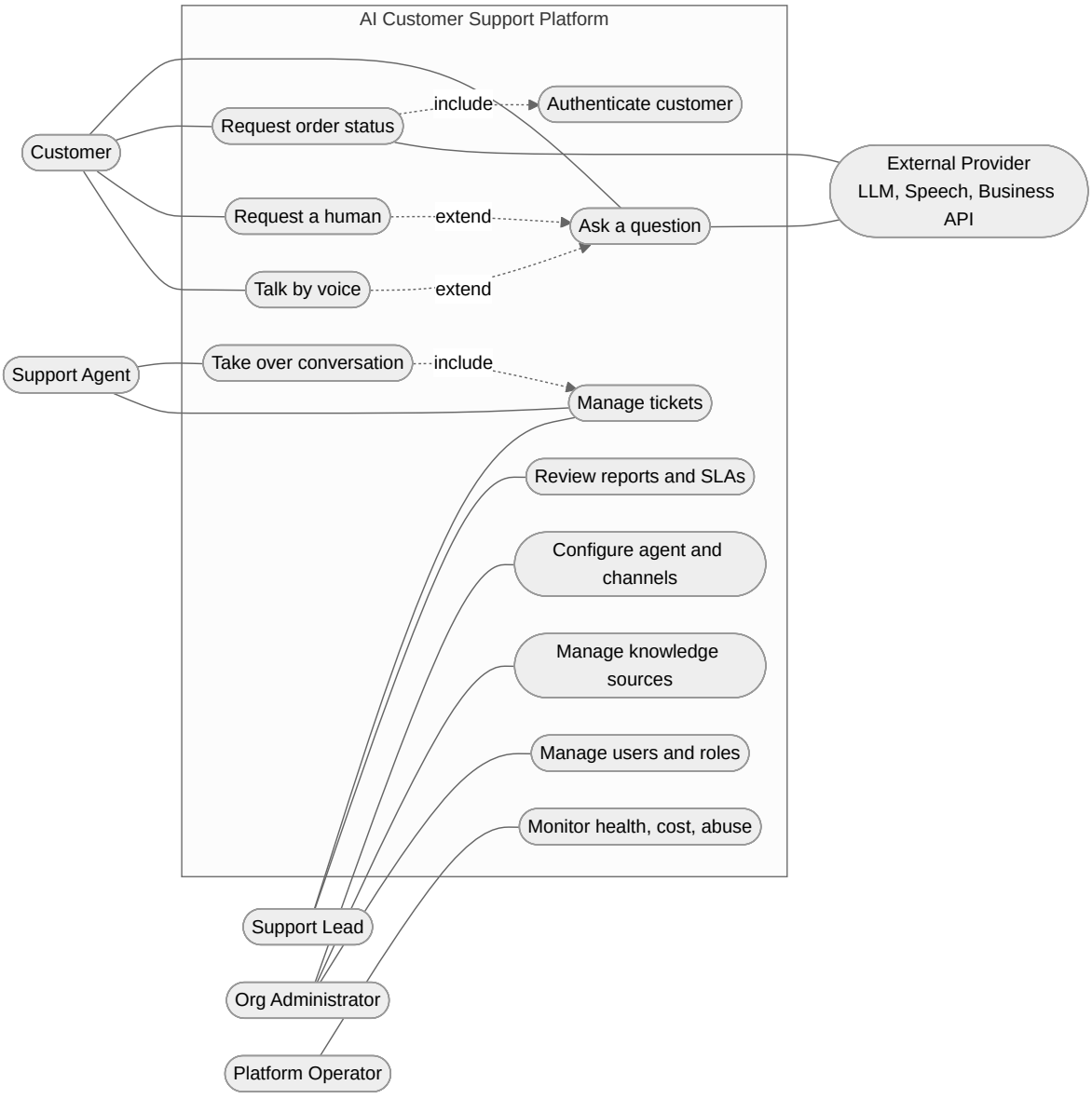


Figure 3.1: System-level use cases

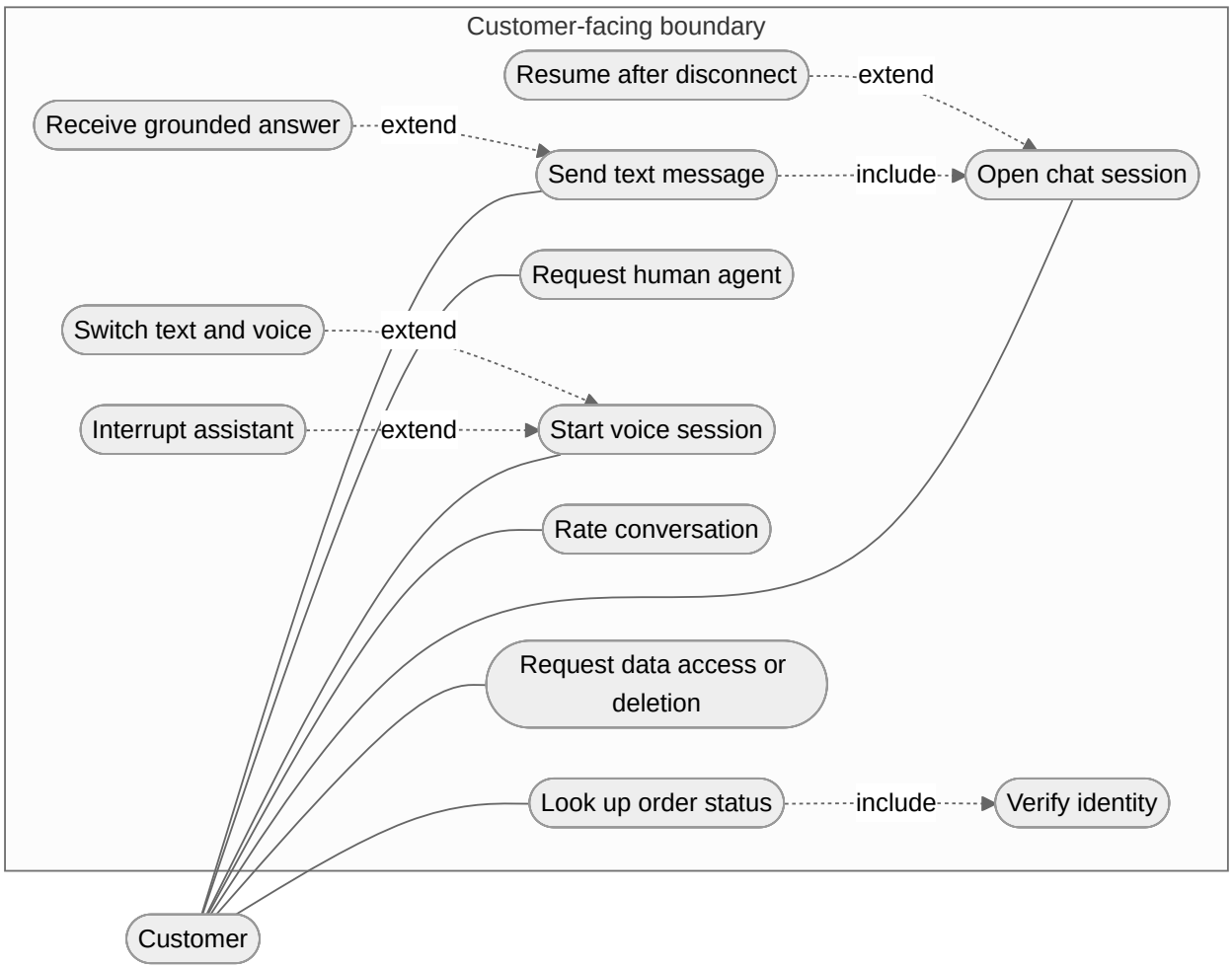


Figure 3.2: Customer use cases

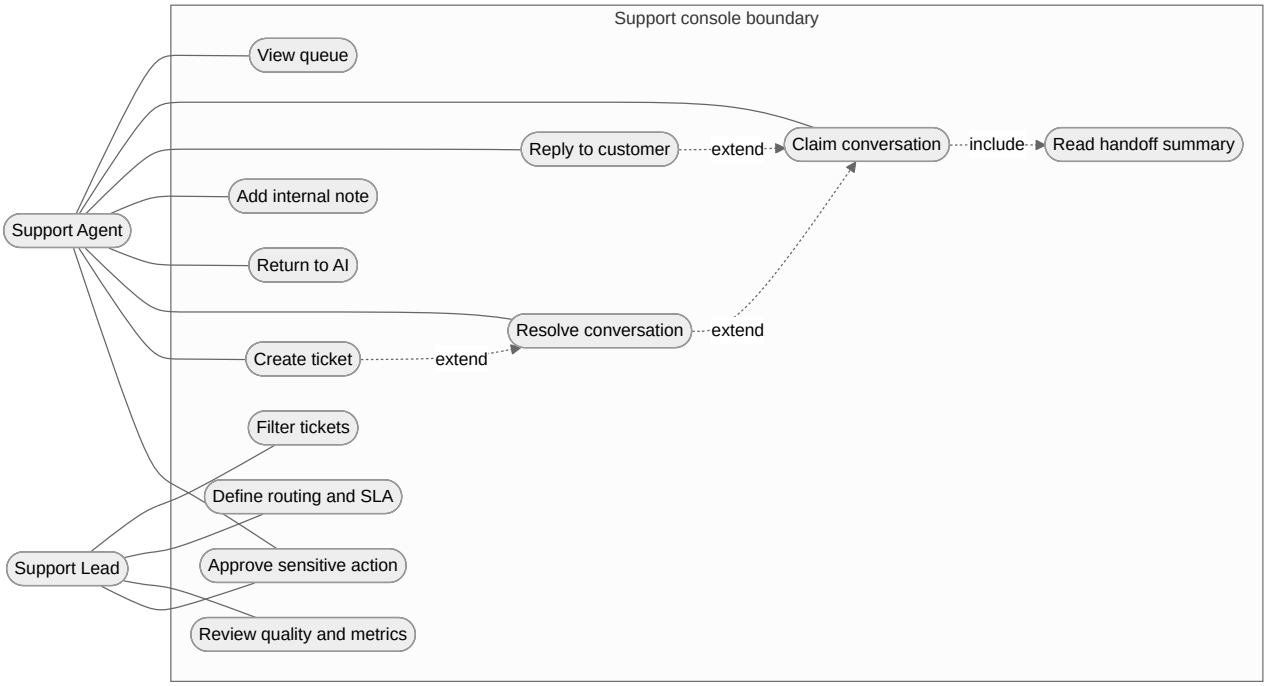


Figure 3.3: Support agent and support lead use cases

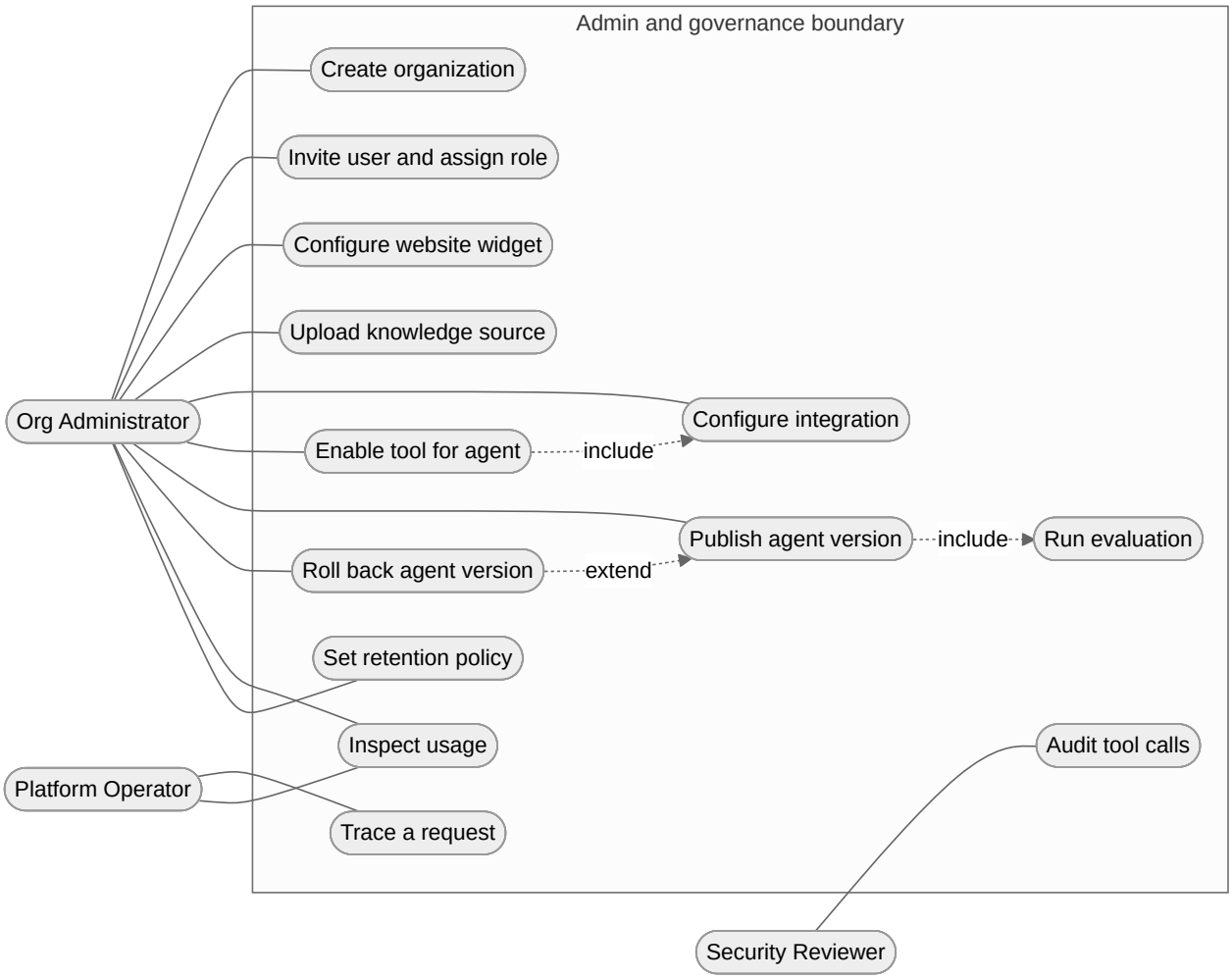


Figure 3.4: Administration and governance use cases

3.4 Actor to epic summary

Table 3.3: Epics mainly used by each actor

Actor	Epics concerned
Customer	E02, E03, E04 (answers), E05, E06 (order status), E07 (human request), E09 (voice), E11 (second channel)
Support agent	E02 (timeline), E07 (queue, handoff, resolution), E08 (tickets), E09 (voice takeover)
Support lead	E08 (filters), E10 (routing, SLA), E12 (metrics)
Administrator	E01, E03 (widget), E04 (sources), E05 (agent versions), E06 (tools), E11, E12 (usage)
Operator	E12, non-functional reliability and security requirements

Chapter 4

Functional overview

4.1 System context

The platform sits between customers, support teams and external systems (models, speech, channels, business systems).

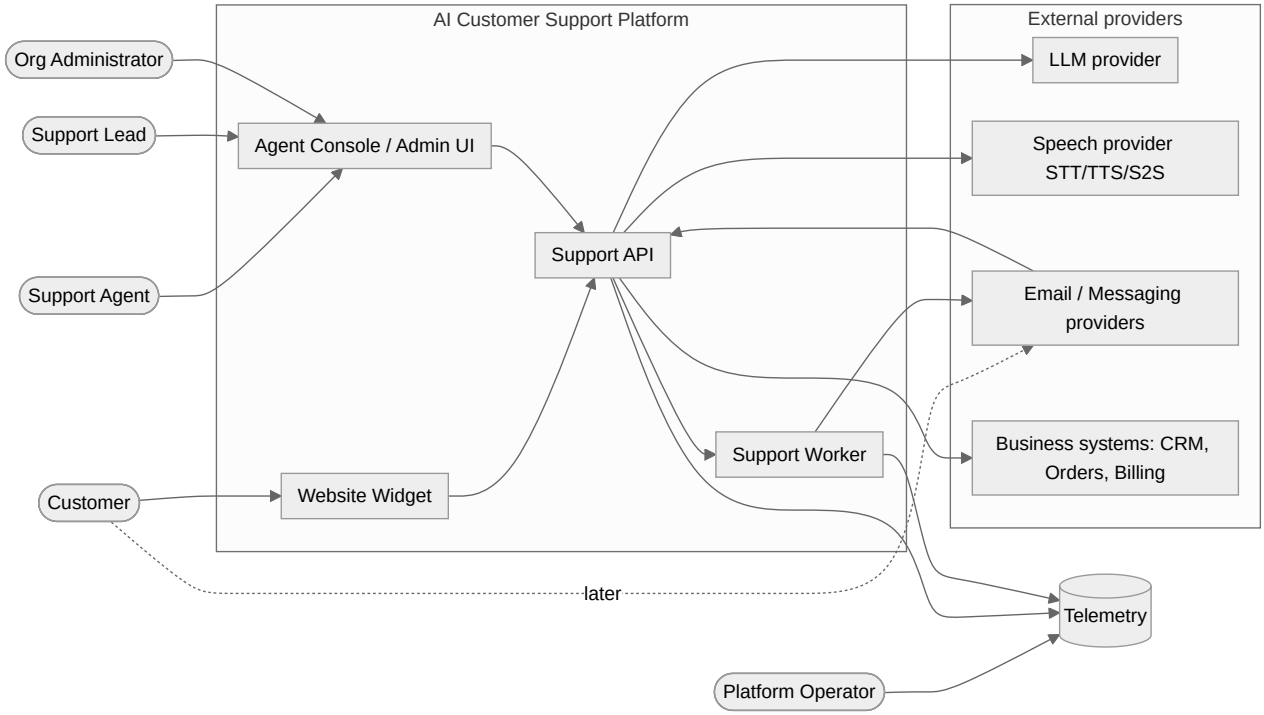


Figure 4.1: System context

4.2 Logical layers

Seven layers organize the system: (1) edge and channel adapters; (2) application API (authentication, tenant resolution, validation, rate limiting); (3) conversation domain; (4) AI orchestration; (5) capability modules (knowledge, business tools, workflow, tickets, human inbox); (6) infrastructure adapters (SQL, object storage, vector index, queue, providers); (7) cross-cutting controls (security, audit, observability, evaluation, usage metering).

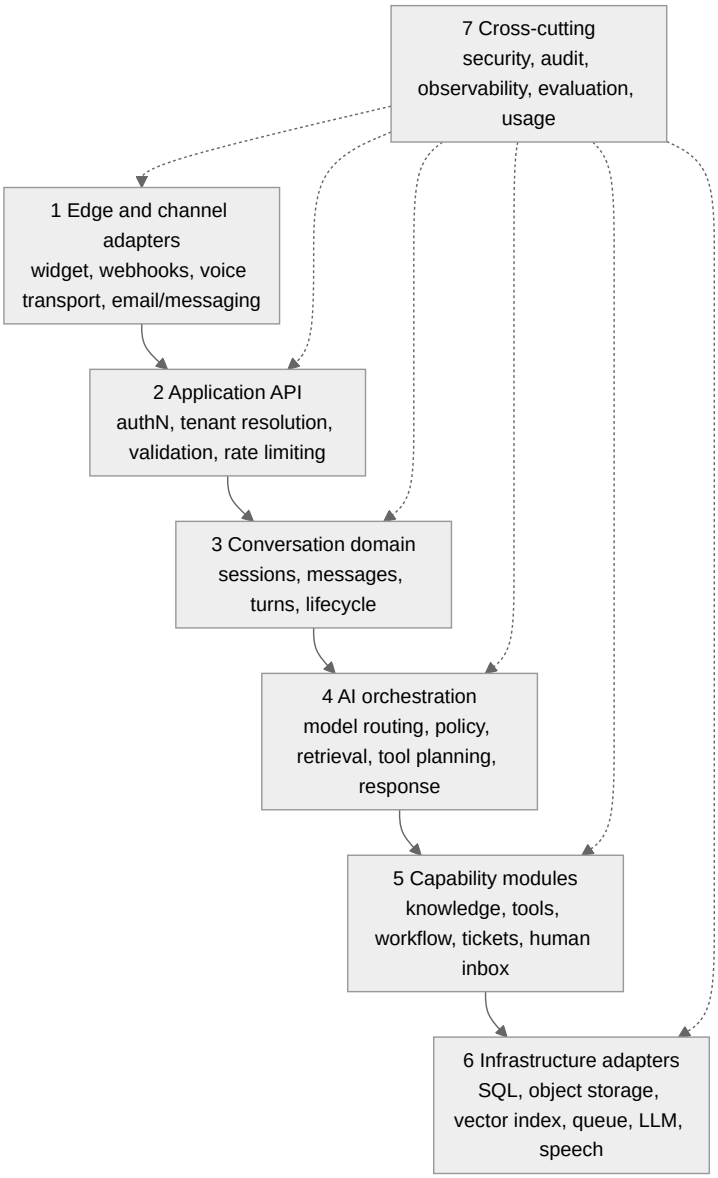


Figure 4.2: Logical layers of the platform

4.3 Functional module catalog

Module names describe logical ownership; they do not imply one deployable service per module.

Table 4.1: Functional modules M01–M20

ID	Module	Main responsibilities
A. Customer interaction and conversation		
M01	Conversation Management	Conversation lifecycle, turns, participants, message ordering, status, links to cases, event history.
M02	Channel Management	Channel connections, provider adapters, webhook intake, outbound delivery, provider identities, retries, delivery state.
M03	Customer Identity and Profiles	Customer profile, contact points, identity linking, preferences, consent records.
B. AI conversation and voice		
M04	AI Conversation Engine	Turn orchestration, intent and context interpretation, policy-aware response planning, model selection, decision trace.
M05	Knowledge Base and Retrieval	Source registration, ingestion, chunking, embeddings, indexing, retrieval, source versions and metadata, citations.
M06	AI Tools and Business Actions	Tool registry, schema validation, permission checks, execution, results, idempotency, approvals, action audit.
M07	Real-Time Voice Sessions	Live audio session lifecycle, streaming transport, turn-taking, interruption handling, timeout, reconnection.
M08	Speech Processing	Speech-provider abstraction, STT, TTS, speech-to-speech, language and voice settings, speech errors.
M09	Transcript and Multimodal Synchronization	Transcript alignment, partial/final state, text and voice correlation, event ordering, modality switching.
C. Support operations and automation		
M10	Human Inbox and Handoff	Agent queues, assignment, human takeover, AI pause/resume, handoff context, internal notes.
M11	Ticket and Case Management	Durable work item, status, priority, category, comments, due dates, resolution.
M12	Workflow Automation	Event-triggered workflows, steps, conditions, schedules, retries, approvals, execution history.
M13	Routing, Policies and SLAs	Routing rules, queue selection, business hours, response and resolution targets, escalation policy, fallback.
M14	Notifications and Follow-Ups	Templates, preferences, dispatch, delivery state, retry.
D. Integrations and insights		
M15	Integration Hub	Connector definitions, credential references, API and webhook mapping, health checks, rate limits, event normalization.
M16	Analytics and Reporting	Operational and product metrics, conversation outcomes, channel usage, SLA reporting, voice latency, cost.
M17	AI Quality, Evaluation and Improvement	Evaluation datasets, test runs, scores, regression comparison, human review, release gates.
E. Platform administration and governance		
M18	Organizations, Users and Access Control	Tenant, memberships, roles, permissions, service identities, invitations, tenant context.

ID	Module	Main responsibilities
M19	Agent Configuration and Administration	Instructions, persona, model/speech configuration references, knowledge assignment, tool allowlist, versions, publication, rollback.
M20	Security, Audit, Observability and Usage	Audit records, retention, abuse controls, telemetry conventions, provider usage, quotas, usage metering.

4.4 Main functional flow (nominal journey)

The following journey describes one text conversation turn in the MVP.

1. The visitor opens the widget: a scoped session is created and the conversation is attached to the right tenant and channel (M02, M01, M18).
2. The visitor sends a message: it is received, deduplicated and durably persisted before acknowledgement (M02, M01).
3. The AI engine resolves trusted context (tenant, conversation, identity, agent version) and checks the conversation state: open, human-owned, blocked or rate-limited (M04).
4. It builds a minimal context, classifies intent and risk, and identifies missing information (M04).
5. When useful, it retrieves approved, tenant-filtered knowledge (M05).
6. It plans the next step using *only* the agent's allowed tools; any proposed tool call is validated, authorized, then executed (M06).
7. It generates a response based on verified results, applies output-safety rules, and persists the reply, sources, tool results, timing, usage and outcome (M04, M01).
8. Delivery, evaluation and analytics events are emitted (M02, M16, M20).
9. If policy requires it or the customer asks, the conversation is handed to a human with its context (M10).

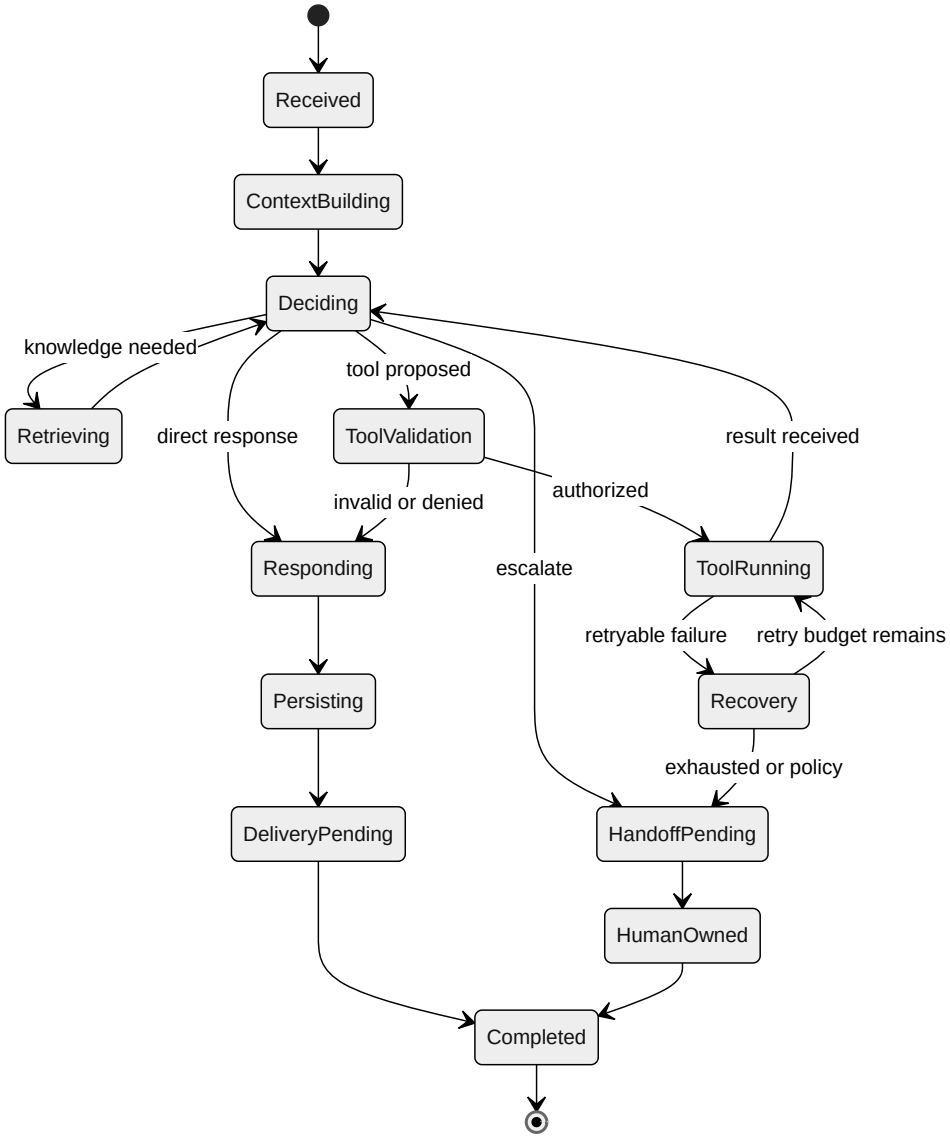


Figure 4.3: States of an AI conversation turn

4.5 Possible engine decisions

At each turn the engine chooses one of the following outcomes:

- answer from approved knowledge;
- ask a clarifying question;
- invoke an allowed business tool;
- create or update a ticket or workflow request;
- refuse or safely redirect a disallowed request;
- hand off to a human;
- recover from a provider or tool failure.

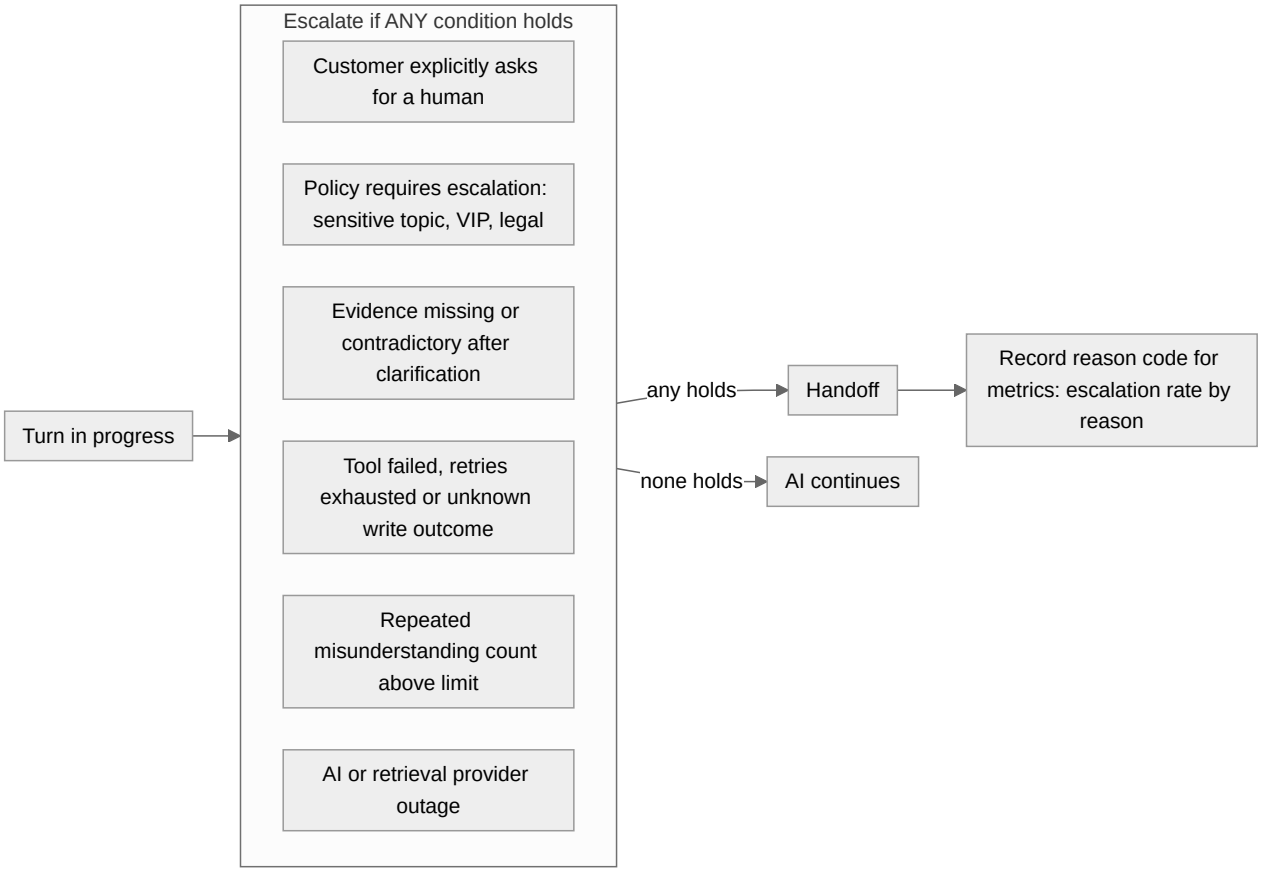


Figure 4.4: Escalation decision tree

4.6 Multimodal conversations

A conversation can contain: customer text messages, finalized speech transcripts, assistant text, generated audio references, system events (voice started or ended) and handoff events. The engine consumes normalized text and intent plus trusted state, without depending on whether the text was typed or transcribed.

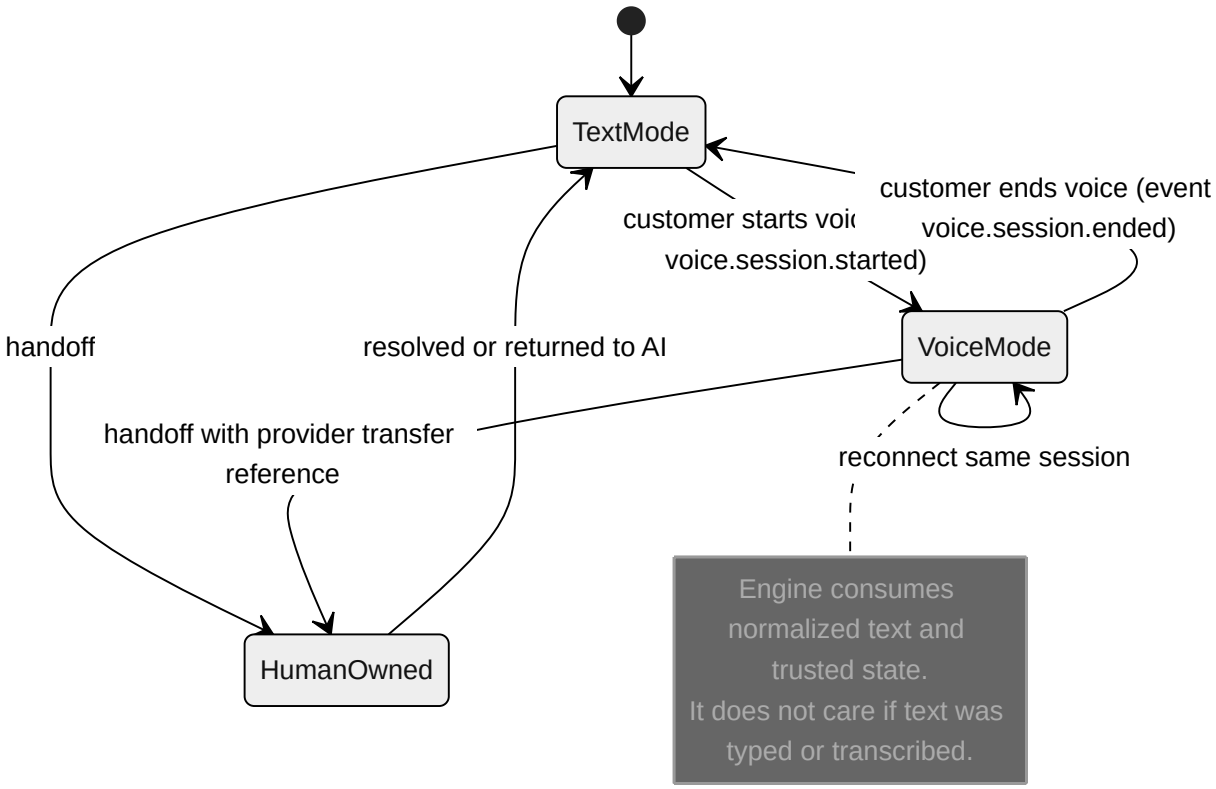


Figure 4.5: Modality switching within one conversation

Chapter 5

Detailed functional specifications

For each of the 12 epics, this chapter gives the objective, the modules concerned, the functional requirements (columns: identifier, statement, MoSCoW priority, release, source in the backlog) and the acceptance criteria. Cross-cutting business rules are grouped in Chapter 6.

5.1 E01 – Tenant and access foundation

Goal: establish secure organization boundaries. **Modules:** M18, M20. **Actors:** administrator, developer or security tester.

ID	Requirement	Prio	Rel.	Source
FR-ACC-01	The system MUST allow an organization to be created and configured: an organization ID is generated, default settings exist and an audit event is recorded.	M	R1	US-001, M18
FR-ACC-02	An administrator MUST be able to invite users; the invitation expires after a defined period and cannot be used afterwards.	M	R1	US-002, M18
FR-ACC-03	An administrator MUST be able to assign roles; the role is validated server-side and a revoked user loses access immediately.	M	R1	US-002, M18
FR-ACC-04	The tenant MUST be resolved from an authenticated identity or a trusted channel configuration, never from a tenant ID supplied in an arbitrary payload.	M	R1	M18, RG-TEN-01
FR-ACC-05	Organization administrator, support agent, customer and service identities MUST be distinct, each with its own permissions.	M	R1	M18
FR-ACC-06	Automated tests MUST demonstrate that any cross-tenant read or write attempt is denied.	M	R1	US-003
FR-ACC-07	Each organization SHOULD be able to configure its channels, default AI agent and handoff policies from a single configuration.	S	R1	US-001, M19

Acceptance criteria. Organization creation records an ID, defaults and an audit event; an expired invitation is refused; an invalid role is rejected by the server; cross-tenant access is denied in CI.

5.2 E02 – Conversation and message core

Goal: persist and retrieve a reliable conversation timeline. **Modules:** M01, M02, M03.

ID	Requirement	Prio	Rel.	Source
FR-CNV-01	A customer MUST be able to start a conversation through website chat; it is scoped to the tenant and channel and its initial event is persisted.	M	R1	US-010
FR-CNV-02	A customer MUST be able to send and receive text messages; message order is stable.	M	R1	US-011
FR-CNV-03	The delivery state of each outbound message (pending, delivered, failed) MUST be represented; errors MUST be recoverable.	M	R1	US-011, M02
FR-CNV-04	A support agent MUST be able to inspect the timeline; messages, system events and visibility levels are distinguished.	M	R1	US-012
FR-CNV-05	A replayed inbound provider event MUST NOT create a duplicate message.	M	R1	US-013
FR-CNV-06	A conversation MUST follow an explicit lifecycle: open, human-owned, closed, reopened.	M	R1	M01
FR-CNV-07	The system MUST support internal notes that are not visible to the customer, with an explicit visibility field.	S	R1	M01, M10
FR-CNV-08	Message kinds MUST be explicit (text, image/file/audio reference, system event, internal note, transcript segment) with dedicated direction and visibility fields.	S	R1	M01

Acceptance criteria. The timeline is identical when re-read; the same provider event delivered twice produces a single message; an internal note is never returned to the customer.

5.3 E03 – Website chat channel

Goal: provide the first customer-facing channel. **Modules:** M02, M19.

ID	Requirement	Prio	Rel.	Source
FR-WEB-01	An administrator MUST be able to configure a website widget associated with the correct organization and agent configuration.	M	R1	US-020
FR-WEB-02	A visitor MUST be able to open the chat and receive a session whose token is scoped, expires and cannot access another tenant.	M	R1	US-021
FR-WEB-03	A visitor MUST be able to resume their conversation after a temporary network failure under a valid session policy.	M	R1	US-022
FR-WEB-04	The widget SHOULD display connection, sending and error states, and who is handling the conversation (AI or human).	S	R1	M02
FR-WEB-05	The system SHOULD rate-limit messages per session to curb abuse.	S	R1	M20

Acceptance criteria. An expired session token is refused; resuming after a disconnection restores the history; an attempt to access another tenant is denied.

5.4 E04 – Knowledge ingestion and RAG

Goal: ground answers in organization-approved sources. **Modules:** M05.

ID	Requirement	Prio	Rel.	Source
FR-KB-01	An administrator MUST be able to register and upload a source; size limits are enforced and the actual content type (not only the extension) is checked.	M	R1	US-030
FR-KB-02	The ingestion state of each source MUST be visible (pending, running, succeeded, failed).	M	R1	US-030
FR-KB-03	The system MUST parse, chunk, embed and index a document asynchronously; a failed ingestion can be retried.	M	R1	US-031
FR-KB-04	Each document MUST be versioned; deleting a source MUST propagate to the index.	M	R1	US-031, M05
FR-KB-05	An AI agent MUST be able to retrieve relevant chunks; results are tenant-filtered and carry source metadata.	M	R1	US-032
FR-KB-06	The customer MUST receive a grounded answer or a clear fallback; an answer is never represented as sourced when retrieval returned no useful evidence.	M	R1	US-033
FR-KB-07	Answers SHOULD indicate the sources used (citations) where supported.	S	R1	M05
FR-KB-08	The system SHOULD detect stale or conflicting sources and trigger a clarification or an escalation.	S	R1	M05, M04
FR-KB-09	Uploaded files SHOULD be malware-scanned where feasible; private objects remain non-public.	S	R1	M05, M20

Acceptance criteria. An uploaded document shows its status and then becomes queryable with source references; a question outside the knowledge gets an explicit fallback with no invented citation; a deleted document is no longer retrievable.

5.5 E05 – AI orchestration

Goal: coordinate each conversation turn safely. **Modules:** M04, M19.

ID	Requirement	Prio	Rel.	Source
FR-IA-01	The system MUST process each turn through a versioned agent configuration; the model/configuration version, latency and outcome are recorded.	M	R1	US-040
FR-IA-02	At each turn the engine MUST choose among: answer from knowledge, clarify, invoke a tool, create/update a ticket, refuse, hand off to a human.	M	R1	M04
FR-IA-03	The customer MUST be able to ask a follow-up question: relevant recent context is used without exposing another customer's data.	M	R1	US-041

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ID	Requirement	Prio	Rel.	Source
FR-IA-04	On a model timeout or provider failure the system MUST apply a bounded retry or fallback, and MUST NEVER deliver a false success message.	M	R1	US-042
FR-IA-05	An administrator MUST be able to publish and roll back an agent version; a published version is immutable and audit-logged.	M	R1	US-043
FR-IA-06	Customer messages, uploaded files, retrieved passages, web pages and tool results MUST be treated as data, never as authoritative instructions.	M	R1	M04, RG-GRD-03
FR-IA-07	The system MUST record concise decision metadata (tools called, sources, outcome) and MUST NOT store hidden reasoning.	M	R1	M04
FR-IA-08	The engine SHOULD detect the customer's language and reply in it; language selection is separate from reasoning.	S	R1	M04
FR-IA-09	The conversation state (open, human-owned, blocked, rate-limited) MUST be checked before any AI processing.	M	R1	M04, M01

Acceptance criteria. Each turn records agent version, duration and outcome; a simulated model failure produces a fallback message or a handoff, never a falsely displayed success; an instruction injected in a retrieved document changes neither policy nor tool permissions.

5.6 E06 – Safe business tools

Goal: let the AI invoke a small allowlisted set of business operations. **Modules:** M06, M15.

ID	Requirement	Prio	Rel.	Source
FR-OUT-01	An administrator MUST be able to enable a versioned tool for an agent; the tool must be allowlisted and its schema valid.	M	R1	US-050
FR-OUT-02	A customer MUST be able to ask for an order status (read-only tool); customer verification and tenant scope are checked; the result is typed and minimal.	M	R1	US-051
FR-OUT-03	Every argument proposed by the model MUST be validated against a schema, and authorization MUST be re-checked at execution time.	M	R1	US-051, M06
FR-OUT-04	On timeout or duplicate requests, idempotency behaviour MUST be defined; an unknown write outcome is not blindly retried.	M	R1	US-052
FR-OUT-05	Every tool call MUST be audited: actor, tool and version, outcome, timestamp, correlation ID, sensitive values redacted.	M	R1	US-053
FR-OUT-06	The system MUST NOT claim that an action succeeded before the tool confirms success.	M	R1	M04, M06

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ID	Requirement	Prio	Rel.	Source
FR-OUT-07	Sensitive or irreversible actions SHOULD require customer confirmation or human approval.	S	R3	M06
FR-OUT-08	Connector failures SHOULD be normalized into categories (authentication, authorization, not found, validation, rate limited, timeout, provider unavailable, conflict, unknown outcome).	S	R1	M06, M15

Acceptance criteria. An order status is returned only to a verified customer of the right tenant; a duplicated call does not produce two executions; the audit trail allows the call to be reconstructed without exposing secrets.

5.7 E07 – Human inbox and handoff

Goal: ensure a human can take over with context. **Modules:** M10, M13.

ID	Requirement	Prio	Rel.	Source
FR-HUM-01	A support agent MUST be able to view an unassigned queue, scoped to the tenant, with assignment.	M	R1	US-060
FR-HUM-02	A customer MUST be able to request a human; the AI stops responding when the handoff policy requires it and the customer receives a clear status.	M	R1	US-061
FR-HUM-03	The system MUST escalate automatically according to policy (missing or conflicting evidence, repeated tool failure, sensitive request, provider outage).	M	R1	M04, M13
FR-HUM-04	The agent MUST receive a handoff summary: transcript, unresolved request, prior tool outcomes permitted by policy, escalation reason.	M	R1	US-062
FR-HUM-05	An agent MUST be able to resolve the conversation; resolution and ownership changes are recorded.	M	R1	US-063
FR-HUM-06	The AI SHOULD be pausable and resumable on a human-owned conversation.	S	R1	M10
FR-HUM-07	The handoff summary MUST NOT silently expose restricted account data.	M	R1	M10, RG-HUM-03

Acceptance criteria. The queue shows only the tenant's conversations; after "I want a human" no AI reply is emitted; the summary contains the transcript and the pending request.

5.8 E08 – Tickets and case tracking

Goal: track support work beyond one interaction. **Modules:** M11.

ID	Requirement	Prio	Rel.	Source
FR-TKT-01	A support agent MUST be able to create a ticket from a conversation; the ticket links to the source conversation and has a status history.	M	R1	US-070
FR-TKT-02	A support lead MUST be able to filter tickets by status, priority and assignment, respecting permissions and tenant boundaries.	M	R1	US-071
FR-TKT-03	A ticket SHOULD carry a category, comments and a due date.	S	R1	M11
FR-TKT-04	A ticket MAY be linked to several conversations (a conversation is an interaction; a ticket is work to resolve).	C	R1	M11

Acceptance criteria. A created ticket can be found from the conversation; the status history is timestamped; a filter never reveals another tenant's tickets.

5.9 E09 – Voice MVP

Goal: support real-time browser voice using the same conversation engine. **Modules:** M07, M08, M09.

ID	Requirement	Prio	Rel.	Source
FR-VOX-01	A customer MUST be able to start and end a voice session; microphone permission is explicit; session state and failures are visible.	M	R2	US-080
FR-VOX-02	A customer MUST be able to speak and receive a spoken response; the speech provider is configurable and failures have a fallback path.	M	R2	US-081
FR-VOX-03	A customer MUST be able to interrupt the AI while it speaks; obsolete audio is cancelled or truncated where supported and is not recorded as fully delivered.	M	R2	US-082
FR-VOX-04	Partial transcripts MUST be provisional; only finalized speech creates a durable turn input; the same audio segment MUST NOT become two independent messages.	M	R2	US-083, M09
FR-VOX-05	A customer MUST be able to switch between text and voice in one conversation; the finalized transcript and message order remain coherent.	M	R2	US-083
FR-VOX-06	A support agent MUST be able to take over a voice conversation or receive its context; the handoff behaviour is explicit for the chosen telephony or media provider.	M	R2	US-084
FR-VOX-07	The system MUST NOT store raw audio by default; audio and transcript retention are defined separately and communicated to the customer.	M	R2	M07, RG-PRV-03
FR-VOX-08	The system SHOULD enforce a maximum session duration and per-tenant voice quotas.	S	R2	M07, M20
FR-VOX-09	The system SHOULD support a modular STT → AI → TTS pipeline first, behind an abstraction that can later host a speech-to-speech provider.	S	R2	M08

Acceptance criteria. An interrupted response is not marked as delivered; turn order is preserved

across typed and spoken messages; a dropped session is cleaned up and reported.

5.10 E10 – Workflow, routing and SLA

Goal: automate repeatable support operations. **Modules:** M12, M13, M14.

ID	Requirement	Prio	Rel.	Source
FR-WKF-01	An administrator MUST be able to define a workflow trigger and steps; workflow version, run state and failures are visible.	S	R3	US-090
FR-WKF-02	A support lead MUST be able to define routing rules and SLA targets; rule priority and fallback behaviour are deterministic.	S	R3	US-091
FR-WKF-03	A failed workflow step MUST be retried safely: the retry policy is bounded and idempotency is defined per step.	S	R3	US-092
FR-WKF-04	The system SHOULD send notifications and follow-ups from templates, with preferences, delivery state and retry; notification delivery is separate from conversational responses.	C	R3	M14
FR-WKF-05	Business-critical procedures MUST be enforced by deterministic workflows; the AI may classify or propose but the workflow enforces the procedure.	S	R3	M12
FR-WKF-06	Business hours and escalation policies SHOULD be configurable per organization.	C	R3	M13

Acceptance criteria. Two rules matching the same conversation always resolve in the same order; a failing step retries within its budget and then lands in a review queue.

5.11 E11 – Integrations and additional channels

Goal: add provider connectors without contaminating core domain logic. **Modules:** M15, M02, M03.

ID	Requirement	Prio	Rel.	Source
FR-INT-01	An administrator MUST be able to configure an integration securely: credentials are secret references and the test status and health are visible.	S	R3	US-100
FR-INT-02	The system MUST normalize a provider webhook into a canonical event: the signature is verified, replays are deduplicated and processing is observable.	S	R3	US-101
FR-INT-03	A customer MUST be able to continue support through a second channel; identity linking follows the verification policy and duplicate conversations are avoided where possible.	S	R3	US-102
FR-INT-04	A phone number or e-mail address MUST NOT, by itself, be treated as proof that a person is authorized to access a protected account.	M	R1	M03, RG-IDN-01

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ID	Requirement	Prio	Rel.	Source
FR-INT-05	The system SHOULD record customer consent and contact preferences.	S	R3	M03
FR-INT-06	Webhook intake MUST enforce body-size and schema limits and respond with success only after durable acceptance.	S	R3	M02, M15

Acceptance criteria. An unsigned or replayed webhook is rejected or deduplicated; the same customer reaching out on two channels is linked only after the required verification.

5.12 E12 – Observability, evaluation and usage

Goal: operate and improve the platform using evidence. **Modules:** M16, M17, M20.

ID	Requirement	Prio	Rel.	Source
FR-OBS-01	An operator MUST be able to trace one request across modules: correlation IDs connect API, worker, model, retrieval and tool activity.	M	R1	US-110
FR-OBS-02	A product owner MUST see resolution and handoff metrics; definitions are documented and metrics are tenant-safe.	S	R1	US-111
FR-OBS-03	The team MUST be able to run AI regression evaluations; the dataset and agent/model version are recorded and critical policy tests gate the release.	M	R1	US-112
FR-OBS-04	An administrator MUST be able to inspect usage by agent and channel; provider usage is reconciled and sensitive data is not exposed in metrics.	S	R3	US-113
FR-OBS-05	The system MUST keep an immutable audit record of governance-relevant events (publication, role changes, tool calls, handoffs).	M	R1	M20
FR-OBS-06	The system SHOULD collect customer feedback after a conversation and link it to the conversation outcome.	S	R3	M16
FR-OBS-07	Voice metrics SHOULD be available: time to first audio, turn latency, interruption rate, reconnect and drop rates.	S	R2	M16, M07

Acceptance criteria. A single trace ID retrieves the activity of one customer message end to end; a release is blocked when a critical safety test fails; metrics contain no message text or personal data in labels.

Chapter 6

Business rules

The following rules apply across epics. Each requirement in Chapter 5 that refers to a rule inherits its constraint.

6.1 Tenant isolation and access

ID	Business rule	Modules
RG-TEN-01	The tenant is resolved from the authenticated identity or from trusted channel configuration; a tenant ID in a customer payload is never trusted.	M18, M02
RG-TEN-02	Tenant filters apply to SQL, search and vector retrieval, object storage, cache keys, jobs, events and analytics.	All
RG-TEN-03	Cross-tenant access tests run in continuous integration; credentials follow least privilege.	M18, M20
RG-ACC-01	Every protected operation checks permissions on the server; the UI alone is never a control.	M18
RG-ACC-02	Authorization is re-checked at execution time, not only when the AI proposes a tool call.	M06

6.2 Identity and verification

ID	Business rule	Modules
RG-IDN-01	A phone number or e-mail address is not proof of authority to access a protected account.	M03
RG-IDN-02	The customer's identity is verified before protected order or account details are exposed.	M03, M06
RG-IDN-03	Identity linking across channels follows the verification policy; duplicate conversations are avoided where possible.	M03, M02

6.3 Conversation and AI behaviour

ID	Business rule	Modules
RG-CNV-01	A conversation is the durable record; channel sessions are delivery mechanisms.	M01
RG-CNV-02	Streaming tokens or audio are not equivalent to a committed final message; only finalized content is persisted as a message.	M01, M09
RG-CNV-03	Durable state is committed before acknowledging a webhook or inbound message when loss would be unacceptable.	M02

ID	Business rule	Modules
RG-GRD-01	The AI does not fabricate order status, account data, policies or actions.	M04
RG-GRD-02	If evidence is missing or contradictory, the AI asks a question or escalates; it distinguishes retrieved policy from live transactional data.	M04, M05
RG-GRD-03	Customer text, documents, retrieved passages and tool results are untrusted data. Instructions embedded in a retrieved document cannot override system policy or tool permissions.	M04
RG-GRD-04	An answer is represented as sourced only when retrieval returned supporting evidence.	M04, M05
RG-GRD-05	The AI does not claim an action succeeded until the tool confirms success.	M04, M06

6.4 Tool execution

ID	Business rule	Modules
RG-TOL-01	Tools are allowlisted per agent version and tenant; tool permissions live outside the model prompt.	M06, M19
RG-TOL-02	Arguments are validated against the input schema before execution; results are validated and returned to the model in minimal, typed form.	M06
RG-TOL-03	Non-idempotent external operations use idempotency keys where supported; an unknown write outcome triggers a status lookup or human review, not a blind retry.	M06
RG-TOL-04	Sensitive or irreversible actions require confirmation or human approval.	M06, M10
RG-TOL-05	Credentials are never exposed to the model; the AI has no unrestricted HTTP access.	M06, M15

6.5 Human handoff

ID	Business rule	Modules
RG-HUM-01	A customer can always request a human; when policy requires, the AI stops responding and the customer sees a clear status.	M10, M04
RG-HUM-02	Escalation triggers include: explicit request, missing or conflicting evidence, repeated tool failure, high-risk request, provider outage.	M04, M13
RG-HUM-03	The handoff bundle contains transcript, concise summary, customer intent, citations used, tool outcomes, ticket and SLA state; it does not expose restricted account data silently.	M10
RG-HUM-04	Resolution and every ownership change are recorded with actor and time.	M10, M01

6.6 Privacy, retention and voice

ID	Business rule	Modules
RG-PRV-01	Personal and sensitive data are classified; retention is defined separately for messages, transcripts, raw audio, uploaded files, logs and audit events.	M20
RG-PRV-02	Unnecessary personal data are kept out of prompts and evaluation datasets; tests use synthetic or redacted data.	M04, M17
RG-PRV-03	Raw audio is not stored by default; the customer is told when voice is active and whether recording or transcription is retained.	M07, M08
RG-PRV-04	Access and deletion workflows are supported where applicable under the law and customer policy.	M20, M03
RG-VOX-01	Partial transcription is provisional; finalized speech creates a durable turn input.	M09
RG-VOX-02	An interrupted assistant utterance is cancelled or truncated and not persisted as fully delivered.	M07, M09

6.7 Reliability

ID	Business rule	Modules
RG-REL-01	Inbound events are stored in a durable inbox; important events use a transactional outbox.	M02, M20
RG-REL-02	External calls have timeouts, bounded retries with backoff and jitter, and idempotency where possible; exhausted retries reach a dead-letter or review queue.	All
RG-REL-03	No database transaction is held open while waiting for a model or external provider.	M04
RG-REL-04	If retrieval is down, the system does not pretend a grounded answer was produced: it asks or hands off. If the AI provider is down, it states temporary unavailability or routes to a human.	M04, M10

Chapter 7

High-level data model

7.1 Core aggregates

Table 7.1: Core aggregates

Aggregate	Content
Organization	Tenant boundary and configuration owner.
Conversation	Durable interaction and message timeline.
Customer	Profile and verified identity links.
Agent configuration / version	Published AI behaviour and allowed capabilities.
Knowledge source / document version	Ingestion and retrieval lifecycle.
Tool execution	Authorization, idempotency, result and audit.
Ticket	Support work item that can span conversations.
Voice session	Live media lifecycle linked to a conversation.
Workflow run	Deterministic multi-step execution.
Audit / usage event	Immutable record for governance and measurement.

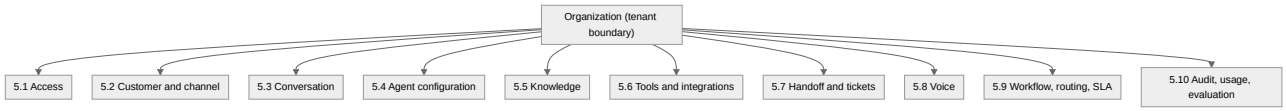


Figure 7.1: Aggregate map

7.2 Key entities per module

Table 7.2: Principal records by module

Mod.	Area	Core records
M01	Conversation	Conversation, ConversationParticipant, ConversationTurn, Message, ConversationEvent
M02	Channels	ChannelConnection, ChannelIdentity, InboundEvent, OutboundDelivery
M03	Customer	Customer, CustomerIdentity, CustomerContact, CustomerConsent
M04	AI engine	AgentDefinition, AgentVersion, AgentExecution, ModelConfiguration
M05	Knowledge	KnowledgeSource, Document, DocumentVersion, DocumentChunk, RetrievalRun
M06	Tools	ToolDefinition, ToolVersion, ToolExecution, ActionApproval
M07	Voice	VoiceSession, VoiceSessionEvent, AudioSegmentReference
M08	Speech	SpeechProvider, SpeechConfiguration, TranscriptionSegment, SpeechGeneration
M09	Transcript	TranscriptSegment, ModalityEvent, MessageMediaReference
M10	Handoff	SupportAgent, SupportQueue, Assignment, HandoffEvent
M11	Tickets	Ticket, TicketComment, TicketStatusHistory
M12	Workflow	WorkflowDefinition, WorkflowVersion, WorkflowRun, WorkflowStepRun
M13	Routing	RoutingRule, SlaPolicy, BusinessSchedule, EscalationPolicy
M14	Notifications	Notification, NotificationTemplate, NotificationDelivery
M15	Integrations	Integration, IntegrationCredentialReference, ConnectorDefinition, IntegrationEvent
M16	Analytics	MetricEvent, ConversationOutcome, Feedback, UsageAggregate
M17	Evaluation	EvaluationDataset, EvaluationRun, EvaluationResult, AgentRelease
M18	Access	Organization, User, Membership, Role, Permission
M19	Agent admin	AgentConfiguration, AgentVersion, AgentDeployment, AgentPolicy
M20	Governance	AuditEvent, UsageRecord, ProviderUsage, RetentionPolicy, SecurityEvent

7.3 Data rules

- All tenant-owned rows include, or are reachable through, a trusted organization ID.
- Internal identifiers are UUIDs or ULIDs; provider IDs are kept in separate fields with uniqueness scoped to the provider connection.
- Phone number or e-mail is never the primary key of a customer.
- Timestamps are stored in UTC and rendered in local time at the edge.
- Raw provider payloads are kept only when needed for debugging or compliance, with explicit retention and redaction.
- Agent configuration, knowledge documents, workflows and tool schemas use immutable version rows.
- Messages, transcripts, audio references and attachments stay distinct so retention and access policies can differ.
- Large binary payloads are stored as object references, not in the primary relational database.
- Deletion and re-indexing propagation of knowledge sources is recorded.
- Hidden chain-of-thought is not stored; only concise decision metadata, tool calls, sources and outcomes needed for debugging and audit.

7.4 Message kinds

Kind	Note
`text`	Customer or assistant text.
`image_reference`	Reference to an approved image.
`file_reference`	Reference to an approved file.
`audio_reference`	Reference to stored audio (only where retention permits).
`system_event`	Lifecycle event (voice started, handoff).
`internal_note`	Not customer-visible.
`transcript_segment`	When transcripts are part of the timeline.

Explicit visibility and direction fields are used rather than conventions in the message text.

Chapter 8

Interfaces

8.1 User interfaces

Three user-facing applications and an operations view are specified functionally here. Visual design is outside the scope of this document.

8.1.1 Customer website widget (E03, E09)

Table 8.1: Widget functions

Function	Behaviour
Open and close chat	Opens a scoped session; shows the welcome state configured by the organization.
Conversation view	Chronological messages with delivery states; distinguishes AI, human agent and system events.
Message composer	Text input with send, error and retry states.
Request a human	Always-available action; displays the resulting status (waiting, connected).
Reconnection	Restores the conversation after a temporary network failure.
Voice controls (R2)	Explicit microphone permission, start and end, active-state indicator, interruption, switch back to text.
Source display	Shows the sources used in a grounded answer where supported.
Consent notice	States whether voice is recorded or transcribed and how long it is retained.

8.1.2 Agent console (E02, E07, E08, E09)

Table 8.2: Agent console functions

Function	Behaviour
Unassigned queue	Tenant-scoped list of conversations awaiting a human, with assignment.
Conversation timeline	Messages, system events and internal notes with distinct visibility.
Handoff summary panel	Transcript, unresolved request, prior tool outcomes, escalation reason.
Take over / release	Pauses or resumes the AI and records ownership changes.
Reply and internal notes	Human replies to the customer; internal notes hidden from the customer.
Ticket actions	Create a ticket from the conversation; follow its status history.
Resolve	Resolves the conversation and records the outcome.
Ticket list	Filters by status, priority and assignment for support leads.

8.1.3 Administration console (E01, E03–E06, E10–E12)

Table 8.3: Administration console functions

Function	Behaviour
Organization settings	Creation, defaults, channel configuration.
Users and roles	Invitations with expiry, role assignment, revocation.
Widget configuration	Link a widget to an organization and an agent configuration.
Knowledge sources	Register and upload sources, view ingestion status, retry, delete.
Agent versions	Edit instructions, assign knowledge and tool allowlist, publish and roll back.
Tools	Enable versioned tools for an agent; view schemas and risk levels.
Integrations	Configure credentials as secret references, run connection tests, view health.
Workflows, routing, SLA (R3)	Define triggers, steps, rules and targets; view run states.
Usage and metrics	Usage by agent and channel; resolution and handoff metrics.

8.1.4 Operations view (E12)

Operators see request, error and latency rates, queue depth and oldest-job age, worker failures and retries, dependency health, model and retrieval signals, and voice signals. They can follow a single trace ID across modules.

8.2 External interfaces

8.2.1 Model provider

Conceptual interface kept behind an adapter:

- `generate_response(input, output_schema, options)`
- `stream_response(input, options)`
- `classify(input, taxonomy, options)`
- `embed(texts, model_configuration)`
- optional `open_realtime_session(configuration)`

Vendor-specific request and response shapes stay in adapters. Model name and version, latency, usage and provider errors are captured without storing secrets.

8.2.2 Speech provider (R2)

Two strategies are supported behind a common abstraction. **Option A**: modular pipeline audio → STT → conversation engine → TTS → audio (explicit transcript, modular provider choice, more latency). **Option B**: speech-to-speech with a real-time multimodal model (potentially more natural, but more provider coupling). One strategy is started first, chosen on latency, language, cost and observability.

8.2.3 Channel providers and webhooks

Webhook handling follows eight steps:

1. verify the provider signature and timestamp;
2. resolve the channel or integration from trusted endpoint configuration;
3. enforce body-size and schema limits;

4. persist the event ID and the payload or minimal normalized data;
5. return success only after durable acceptance;
6. process asynchronously;
7. deduplicate provider retries;
8. record processing state and failure reason without leaking secrets.

8.2.4 Business tools and connectors

Tool integration has three layers: the *tool definition* (what the AI may request), the *tool execution service* (validation, authorization, idempotency, audit, timeout, result normalization) and the *connector adapter* (provider-specific authentication and mapping). The AI never receives raw credentials.

Table 8.4: Fields of a tool definition

Field	Description
Stable name and description	For example `orders.lookup`.
Version	Immutable once published.
Input and output schemas	JSON schemas with `additionalProperties` disallowed.
Allowed scope	Agents and tenants permitted to use the tool.
Required permission	Checked by code at execution.
Risk level	Read-only, reversible write, or sensitive write.
Confirmation / approval	Whether customer confirmation or human approval is required.
Timeout and retry policy	Bounded; safe-to-retry only.
Idempotency behaviour	How duplicates are handled.
Audit and redaction policy	Which fields are redacted.

Example contract for the MVP read-only tool (illustrative, not a claim about any commerce provider):

`orders.lookup`, version 1, risk: read-only

Input: `order_reference` (string, 3 to 80 characters), required, no additional properties.
Output: `status` (string) and `estimated_delivery` (string or null), both required, no additional properties.

Connector failures are mapped to the categories: `AUTHENTICATION_FAILED`, `AUTHORIZATION_DENIED`, `NOT_FOUND`, `VALIDATION_FAILED`, `RATE_LIMITED`, `TIMEOUT`, `PROVIDER_UNAVAILABLE`, `CONFLICT`, `UNKNOWN_OUTCOME`. Only safe-to-retry failures are retried.

8.3 Internal event contracts

Modules communicate through synchronous interfaces and asynchronous events. Every event and request carries `trace_id`, `correlation_id`, `causation_id`, `tenant_id`, and where relevant `conversation_id`, `turn_id`, `tool_execution_id` and `provider_request_id`. Message text, e-mail, phone numbers and tokens never appear in metric labels.

Chapter 9

Non-functional requirements

Numeric targets are **TBD**: they are set after a measured baseline and agreed with stakeholders. Qualitative requirements below are mandatory unless stated otherwise.

9.1 Security

ID	Requirement	Prio	Rel.	Source
NFR-SEC-01	Tenant isolation MUST be enforced in application authorization and data access (SQL, search, objects, caches, jobs, events, analytics), not only in the UI.	M	R1	RG-TEN-*
NFR-SEC-02	Provider credentials MUST be stored in a secret manager or protected encrypted store and referenced by identifier; secrets MUST be redacted from logs and traces and support rotation and revocation.	M	R1	M15, M20
NFR-SEC-03	Customer text, uploads, retrieved passages and tool results MUST be treated as untrusted; tool permissions MUST live outside the prompt; arguments MUST be validated.	M	R1	RG-GRD-03
NFR-SEC-04	Uploads MUST be limited in size and type, validated on content, kept private by default, and served through short-lived signed URLs when needed.	M	R1	FR-KB-01
NFR-SEC-05	Webhooks MUST verify signature and timestamp, resist replay, and be deduplicated.	M	R3	FR-INT-02
NFR-SEC-06	The system SHOULD apply rate limits and abuse controls at the edge and per session.	S	R1	FR-WEB-05
NFR-SEC-07	The security test matrix (Section 11.4) MUST pass before each release.	M	R1	Chap. 11

9.2 Privacy and compliance

ID	Requirement	Prio	Rel.	Source
NFR-PRV-01	Personal and sensitive data MUST be classified; retention MUST be defined separately for messages, transcripts, raw audio, uploads, logs and audit events.	M	R1	RG-PRV-01
NFR-PRV-02	Access to transcripts and recordings MUST be limited by role.	M	R1	M18, M20
NFR-PRV-03	The system MUST support access and deletion workflows where applicable under law and customer policy.	S	R1	RG-PRV-04

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ID	Requirement	Prio	Rel.	Source
NFR-PRV-04	Voice use MUST be preceded by explicit microphone permission and a clear notice of any retention.	M	R2	FR-VOX-01, FR-VOX-07
NFR-PRV-05	Synthetic or redacted data SHOULD be used for tests and evaluation datasets.	S	R1	RG-PRV-02

9.3 Reliability and availability

ID	Requirement	Prio	Rel.	Source
NFR-REL-01	Inbound events MUST be recorded in a durable inbox and important events published through a transactional outbox.	M	R1	RG-REL-01
NFR-REL-02	External calls MUST use timeouts, bounded retries with exponential backoff and jitter, idempotency keys, and circuit breakers.	M	R1	RG-REL-02
NFR-REL-03	The system MUST degrade gracefully: it never fakes a grounded answer when retrieval is down, and routes to a human or communicates unavailability when the model provider is down.	M	R1	RG-REL-04
NFR-REL-04	Health checks and dependency readiness MUST be exposed; exhausted retries MUST reach a review queue.	M	R1	M20
NFR-REL-05	Availability, message-delivery success and recovery objectives are **TBD**; backup and restore MUST be exercised before the R1 release.	M	R1	Sprint 6

9.4 Performance

ID	Requirement	Prio	Rel.	Source
NFR-PERF-01	Median and p95 text response latency MUST be measured end to end; targets are TBD.	M	R1	FR-OBS-02
NFR-PERF-02	Voice latency (time to first audio, end-of-speech to response start, total turn latency) MUST be measured; targets are TBD.	M	R2	FR-OBS-07
NFR-PERF-03	Model, retrieval and tool budgets (timeouts, token and cost caps) MUST be defined per agent version.	S	R1	M04, M19
NFR-PERF-04	Ingestion MUST run asynchronously so that uploads never block customer conversations.	M	R1	FR-KB-03

9.5 Observability

ID	Requirement	Prio	Rel.	Source
NFR-OBS-01	Every request and job MUST carry a trace ID and tenant context; correlation IDs connect API, worker, model, retrieval and tool activity.	M	R1	FR-OBS-01
NFR-OBS-02	Metric labels MUST avoid sensitive and high-cardinality values (message text, e-mail, phone, tokens).	M	R1	Chap. 8
NFR-OBS-03	Operational signals MUST cover API, queues, workers, database, conversation delivery, AI and RAG, and voice.	M	R1	M16, M20

9.6 Quality of AI responses

ID	Requirement	Prio	Rel.	Source
NFR-AIQ-01	A curated evaluation set MUST cover FAQ, ambiguity, missing information, stale or conflicting documents, prompt injection, restricted data, tool success and failure, provider timeout, human request, language switching, and voice interruption.	M	R1	FR-OBS-03
NFR-AIQ-02	Evaluation MUST score correctness, groundedness, clarification, tool choice and arguments, authorization compliance, handoff decisions, tone, latency and cost.	M	R1	M17
NFR-AIQ-03	Deterministic assertions MUST be used for permissions, schemas and state transitions; model-based judges are only one signal, never the sole release gate.	M	R1	M17
NFR-AIQ-04	A new prompt, model or agent version MUST NOT be published until critical safety and tool-authorization tests pass, latency and cost are measured, a rollback path is known and a versioned release record exists.	M	R1	FR-IA-05

9.7 Maintainability, portability and usability

ID	Requirement	Prio	Rel.	Source
NFR-MNT-01	The platform MUST be a modular monolith with strict module boundaries; services are extracted only when justified.	M	R1	ADR-001
NFR-MNT-02	Model, speech, channel and integration providers MUST be reachable only through adapters, verified by contract tests.	M	R1	RG, M04, M15
NFR-MNT-03	Database migrations MUST be reviewed and repeatable; public API contracts MUST be documented and versioned.	M	R1	DoD
NFR-USA-01	The widget SHOULD be usable on desktop and mobile browsers; accessibility and supported languages are to be specified with the client (TBD).	S	R1	M02

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ID	Requirement	Prio	Rel.	Source
NFR-USA-02	Customer-facing responses SHOULD be concise and understandable; text and audio rendering are separate from reasoning.	S	R1	M04

Chapter 10

Planning, constraints and risks

10.1 Releases

Table 10.1: Release goals

Rel.	Name	Expected outcome
R1	Text support MVP	A website visitor receives knowledge-grounded answers, one read-only business lookup is available, and human handoff preserves context.
R2	Voice MVP	A customer can speak with the same AI engine and see a coherent transcript.
R3	Operational expansion	Support teams can scale operations with workflows, routing, additional channels and stronger evaluation.

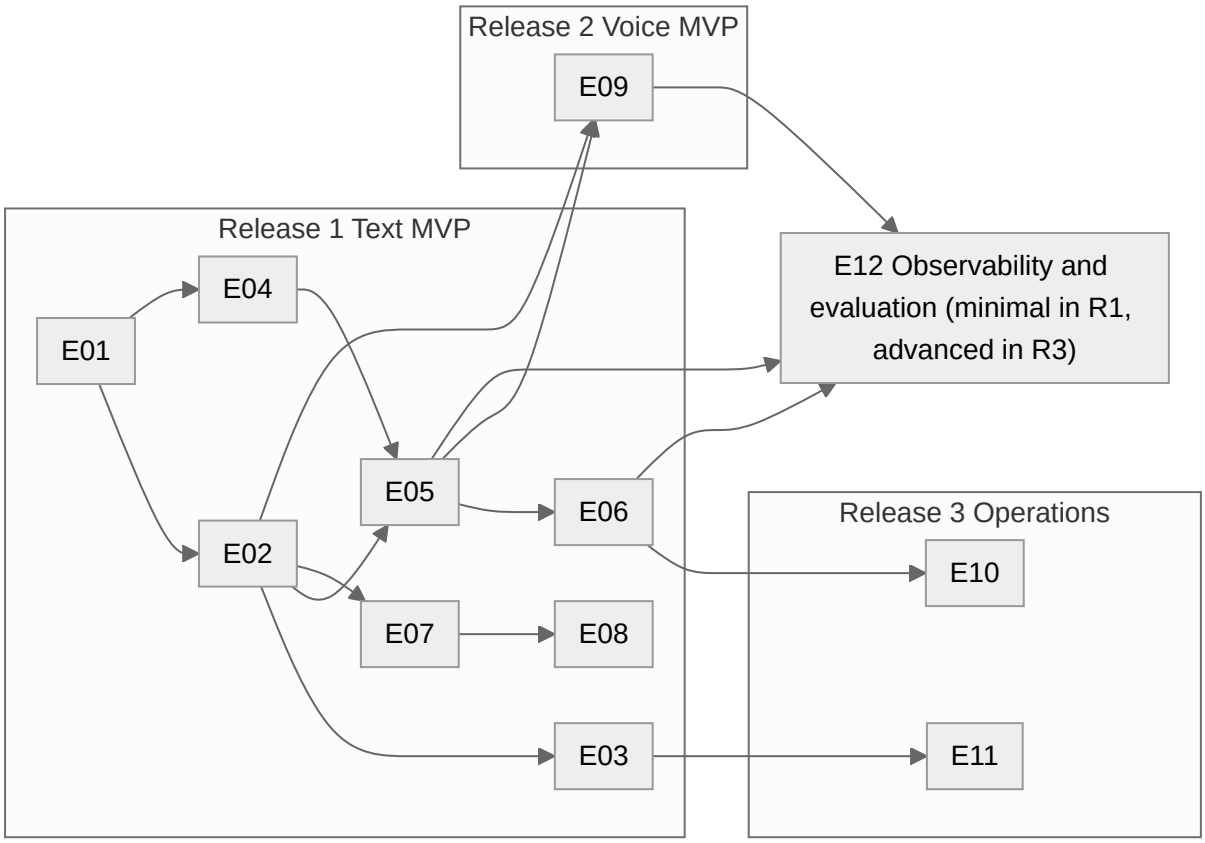


Figure 10.1: Epic delivery map by release

10.1.1 R1 exit criteria

- tenant-isolation tests pass;
- inbound messages are deduplicated;

- unknown or unsupported questions have a safe fallback;
- tool calls are allowlisted, authorized and audited;
- a human can take over;
- core API and worker telemetry is available;
- the evaluation suite has no unresolved critical policy failure.

10.1.2 R2 exit criteria

- session start, stop and cleanup are reliable;
- transcript finalization and ordering work;
- interruption behaviour is tested;
- the audio and transcript privacy policy is implemented;
- latency and cost metrics are visible;
- human handoff behaviour is validated for the selected provider.

10.2 MoSCoW summary

Table 10.2: Requirements by release and priority

Epic	Must	Should	Could	Release
E01 Tenant and access foundation	6	1	0	R1
E02 Conversation and message core	6	2	0	R1
E03 Website chat channel	3	2	0	R1
E04 Knowledge ingestion and RAG	6	3	0	R1
E05 AI orchestration	8	1	0	R1
E06 Safe business tools	6	2	0	R1–R3
E07 Human inbox and handoff	6	1	0	R1
E08 Tickets and case tracking	2	1	1	R1
E09 Voice MVP	7	2	0	R2
E10 Workflow, routing and SLA	0	4	2	R3
E11 Integrations and additional channels	1	5	0	R1–R3
E12 Observability, evaluation and usage	3	4	0	R1–R2–R3
Total	**54**	**28**	**3**	

The counts include the functional requirements of Chapter 5 only; non-functional requirements are tracked in Chapter 9.

10.3 Provisional Sprint plan

Sprints are assumed to last two weeks. The sequence is a planning hypothesis, to be re-forecast after the first Sprint using actual capacity.

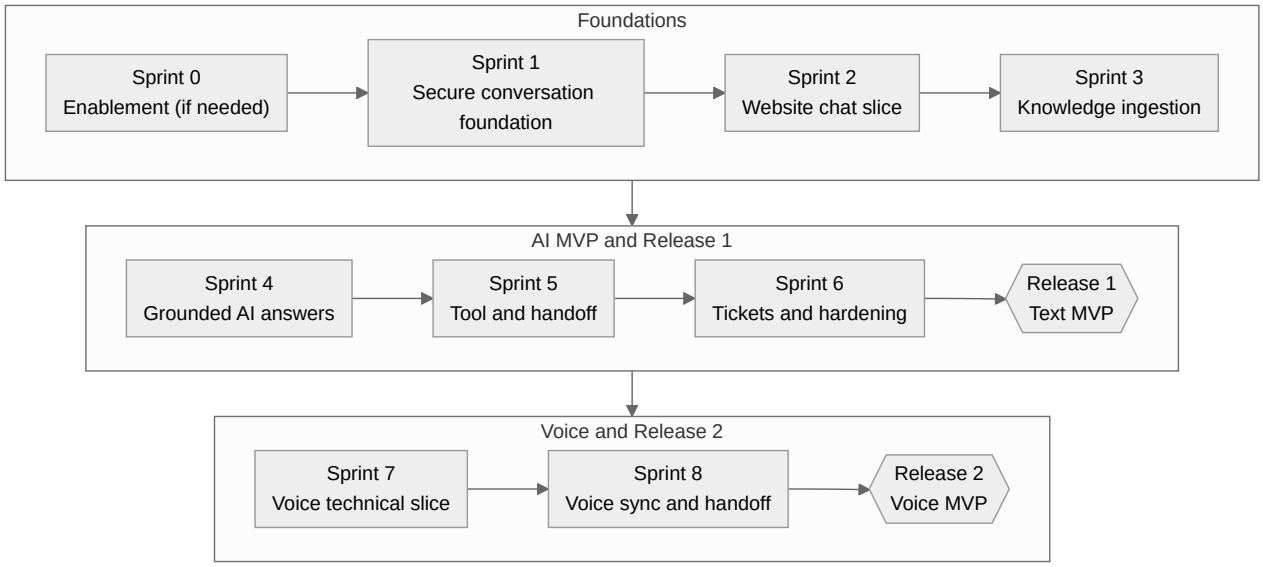


Figure 10.2: Sprint sequence and goals

Table 10.3: Provisional Sprints and related requirements

Sprint	Goal	Stories and main requirements
Sprint 0 (if needed)	Make the team able to build, test and release safely.	Repository and module structure, CI, local environment, migrations, initial threat model, ADRs, minimal deployment and secret handling, Definition of Done.
Sprint 1	A tenant-scoped conversation and message can be created, persisted and retrieved.	US-001, US-003, US-010, US-011; FR-ACC-01, FR-ACC-06, FR-CNV-01, FR-CNV-02.
Sprint 2	A visitor can exchange text messages with the backend reliably.	US-020, US-021, US-022, US-013; FR-WEB-01..03, FR-CNV-05.
Sprint 3	Organization knowledge can be ingested and retrieved with tenant boundaries.	US-030, US-031, US-032; FR-KB-01..05.
Sprint 4	Website chat answers common questions from approved knowledge and handles missing evidence safely.	US-040, US-033, US-041, US-042; FR-IA-01..04, FR-KB-06.
Sprint 5	The AI performs one authorized read-only lookup and hands off unresolved conversations.	US-050, US-051, US-053, US-060, US-061, US-062; FR-OUT-01..05, FR-HUM-01..04.
Sprint 6	Support staff track unresolved work and the MVP meets release gates.	US-070, US-071, US-043, US-110, US-112; FR-TKT-01..02, FR-IA-05, FR-OBS-01, FR-OBS-03.
Sprint 7	Validate the voice architecture and complete a minimal browser voice turn.	US-080, US-081; FR-VOX-01..02, voice telemetry, consent review.
Sprint 8	Voice and text share a coherent conversation timeline.	US-082, US-083, US-084; FR-VOX-03..06.
Later	Additional channels, workflow, routing, SLA, evaluation, usage plans.	E10, E11, US-113; FR-WKF, FR-INT, FR-OBS-04.

10.4 Constraints

- **Architecture:** modular monolith first; the core domain does not depend on any specific provider.
- **Security:** AI output is untrusted; authorization is enforced in code.
- **Privacy:** no raw audio or sensitive data retained without need, consent and policy.
- **Organization:** small team; capacity and calendar unknown; dates are placeholders.
- **Dependencies:** voice and integration work depend on vendor capabilities and credentials.

10.5 Initial risks

Table 10.4: Risks and mitigations

Risk	Impact	Mitigation
Prompt injection via customer or retrieved content	Unauthorized or incorrect actions	Tool allowlist, server authorization, untrusted-content boundaries, adversarial tests.
Tenant data leakage in retrieval	Severe privacy and security issue	Tenant-scoped indexes and filters, access tests, per-tenant metadata and object controls.
LLM latency and cost	Poor customer experience, high cost	Budgets, timeouts, model routing, caching only safe content, measurement.
Duplicate webhook or tool execution	Duplicate messages or actions	Durable inbox, idempotency keys, deduplication.
Voice interruption and transcript race	Confusing conversation history	Sequence IDs, finalized versus partial states, cancellation semantics.
Provider lock-in	Cost and migration risk	Provider adapters, normalized contracts, contract tests.
Unsupported AI answer	Loss of trust	Retrieval evaluation, evidence thresholds, safe fallback and handoff.
Scope too broad	Delayed first usable product	Text MVP vertical slice, ordered backlog, defer non-essential channels.

10.6 Decisions to record and to defer

Architecture decision records to create: ADR-001 modular monolith boundaries; ADR-002 relational database and tenant isolation; ADR-003 queue and outbox/inbox; ADR-004 knowledge ingestion and vector search; ADR-005 LLM provider abstraction and data policy; ADR-006 voice strategy (modular STT/TTS versus speech-to-speech); ADR-007 browser voice transport and telephony; ADR-008 transcript and audio retention and consent; ADR-009 authentication and customer verification; ADR-010 tool authorization and approval.

Decisions not to make prematurely: one microservice per module; a universal workflow builder before core support works; several speech providers before the first voice use case is validated; autonomous write access to every customer system; permanent raw audio retention; complex billing before pricing and usage are known.

Chapter 11

Acceptance and validation

11.1 Acceptance principles

- A requirement is accepted when its acceptance criteria (Chapter 5) are demonstrated and the relevant release exit criteria (Chapter 10) are met.
- Each Sprint ends with a Sprint Review where the Product Owner inspects a working Increment, not only design documents.
- Acceptance is evidence-based: automated tests, an evaluation run, a trace, or a demonstration on a test tenant with synthetic data.

11.2 Definition of Ready

A backlog item is ready for Sprint Planning when: the user or business outcome is clear; acceptance criteria are testable; dependencies and provider assumptions are identified; UX, API and data behaviour are sufficiently understood; security and privacy implications have been considered; the item fits in one Sprint; an estimate or sizing approach exists; and the required credentials, test data and environments are available or planned.

11.3 Definition of Done

An item is Done only when:

- the implementation meets its acceptance criteria;
- code is reviewed and respects module boundaries;
- automated tests cover relevant behaviour and failure cases;
- tenant isolation and authorization are tested where relevant;
- input validation and error handling are implemented;
- database migrations are reviewed and repeatable;
- logs, metrics and traces include safe correlation fields and redact sensitive values;
- documentation and API contracts are updated;
- no known critical or high-severity defect remains without explicit risk acceptance;
- deployment, rollback or feature-flag behaviour is understood;
- the Product Owner can inspect the completed behaviour.

11.3.1 AI-specific checks

Model and prompt or configuration versions are identifiable; tool schemas are validated; tool permissions are enforced in code; retrieval is tenant-scoped; failure, empty-retrieval and uncertainty paths are tested; a regression example is added for important behaviour; and no secrets or hidden reasoning are persisted in logs.

11.3.2 Voice-specific checks

Microphone permission and active state are clear; session cleanup works on disconnect; interruption and cancellation are tested; transcript finalization and ordering are tested; audio and transcript retention follow policy; latency and failure metrics are emitted.

11.4 Security test matrix

Table 11.1: Security and robustness scenarios to pass before a release

Scenario	Expected result
Cross-tenant read and write	Denied; no data returned.
Unauthorized tool invocation	Refused; audited.
Prompt injection through user input and retrieved documents	Policy and tool permissions unchanged.
Webhook signature failure and replay	Rejected or deduplicated.
Duplicate webhook delivery	One message only.
Tool timeout with unknown write outcome	No blind retry; status check or human review.
Secret leakage in logs and traces	No secret present.
Unauthorized access to transcripts or audio	Denied.
Rate-limit and abuse scenarios	Throttled; service remains available.

11.5 Acceptance per release

Table 11.2: Acceptance activities per release

Rel.	Evidence required
R1	Passing tenant-isolation and security matrix; end-to-end demo (question → grounded answer → order lookup → handoff with context); evaluation report without critical failure; audit, trace and metrics shown; rollback path demonstrated.
R2	Voice demo (speak, interrupt, type a follow-up, hand off); transcript ordering tests; privacy and consent review; latency and cost dashboard; voice regression scenarios.
R3	Workflow, routing and SLA determinism tests; second-channel identity-linking tests; usage reconciliation; advanced evaluation gates.

11.6 Validation of requirements

Requirements are reviewed with the stakeholders listed in Table 2.2. Each change is traced in the revision history. Open items marked TBD are listed in Appendix E.

Appendix A

Glossary

ADR Architecture Decision Record.

Agent (AI agent) a versioned configuration of instructions, knowledge assignments, allowed tools and policies.

Agent (support agent) a human member of the support team.

Barge-in the customer interrupting the AI while it is speaking.

Chunk a segment of a document prepared for embedding and retrieval.

Connector a provider-specific adapter that calls an external business system.

Correlation ID an identifier that links all activity caused by one request across modules.

Embedding a numeric representation of text used for semantic retrieval.

Grounding basing an answer on retrieved, approved evidence.

Handoff transfer of a conversation from the AI to a human, with context.

Idempotency key a value that lets a retried command be applied at most once.

Inbox / outbox durable stores for received events and for events to publish reliably.

Increment a usable, potentially releasable result of a Sprint.

MoSCoW Must, Should, Could, Won't prioritization.

MVP minimum viable product; here release R1.

RAG retrieval-augmented generation.

SLA service-level agreement: response and resolution targets.

STT / TTS speech-to-text and text-to-speech.

Tenant a customer organization; the data-isolation boundary.

Tool an allowlisted, schema-defined business operation the AI may request.

Turn the processing of one customer message.

Webhook an HTTP callback sent by a provider to the platform.

Appendix B

Detailed scenarios

B.1 Scenario 1 – Grounded answer on website chat

Field	Description
Actors	Customer, AI engine.
Preconditions	The organization has an agent version published, a widget configured and at least one ingested knowledge source.
Trigger	The customer opens the widget and asks a question.
Main flow	1. A scoped session starts and a conversation is created. 2. The message is persisted and deduplicated. 3. The engine checks state, builds context and retrieves tenant-filtered passages. 4. It answers using the evidence and cites the sources. 5. The answer, sources, timing and usage are persisted and delivered.
Alternative flows	*No useful evidence:* the engine asks a clarification or gives a clear fallback and offers a human; it does not present the answer as sourced. *Model timeout:* bounded retry, then fallback message or handoff.
Postconditions	Conversation timeline updated; audit and trace recorded.
Requirements	FR-CNV-01/02, FR-KB-05/06, FR-IA-01..04.

B.2 Scenario 2 – Order status lookup

Field	Description
Actors	Customer, AI engine, tool execution service, commerce system.
Preconditions	The `orders.lookup` tool (read-only) is enabled for the agent; the customer can be verified.
Trigger	The customer asks where their order is.
Main flow	1. The engine proposes `orders.lookup` with an order reference. 2. The system validates the schema, tenant scope and customer verification. 3. The connector queries the commerce system. 4. A typed, minimal result returns. 5. The engine answers from the verified result. 6. The call is audited with sensitive values redacted.
Alternative flows	*Verification fails:* no data is exposed; the engine asks for verification or offers a human. *Timeout or unknown outcome:* no blind retry; the customer is told the status is unavailable, or the case is handed off.
Postconditions	Audit record with actor, tool version, outcome, timestamp and correlation ID.
Requirements	FR-OUT-01..06, RG-IDN-02, RG-TOL-01..03.

B.3 Scenario 3 – Customer requests a human

Field	Description
Actors	Customer, AI engine, support agent.
Preconditions	A conversation is open.
Trigger	The customer asks for a human, or policy triggers escalation.
Main flow	1. The conversation moves to a pending-handoff state and the AI stops responding. 2. The customer sees a clear status. 3. The conversation enters the tenant's unassigned queue with the handoff summary. 4. An agent claims it and takes ownership. 5. The agent replies, optionally creates a ticket, then resolves.
Alternative flows	*No agent available:* the customer is told so and a ticket is created for follow-up.
Postconditions	Ownership changes and resolution recorded; transcript preserved.
Requirements	FR-HUM-01..07, FR-TKT-01.

B.4 Scenario 4 – Knowledge document ingestion

Field	Description
Actors	Administrator, ingestion worker.
Trigger	The administrator uploads a document.
Main flow	1. Size and content type are validated. 2. A source and a document version are registered with state *pending*. 3. The worker parses, chunks, embeds and indexes. 4. The state moves to *succeeded*; the document is retrievable with source metadata.
Alternative flows	*Ingestion fails:* the state shows the failure and the job can be retried. *Document deleted:* the deletion propagates to the index.
Requirements	FR-KB-01..04.

B.5 Scenario 5 – Voice turn with interruption (R2)

Field	Description
Actors	Customer, speech provider, AI engine.
Trigger	The customer starts a voice session and speaks.
Main flow	1. Microphone permission is requested explicitly. 2. Partial transcripts are shown as provisional. 3. The final transcript creates a durable turn input. 4. The engine answers; speech is synthesized and played. 5. If the customer interrupts, the obsolete audio is cancelled or truncated and not recorded as fully delivered. 6. The customer may switch to text; the timeline stays coherent.
Postconditions	Finalized transcript and events linked to the conversation; no raw audio kept by default.
Requirements	FR-VOX-01..07, RG-VOX-01/02.

Appendix C

Traceability matrix

Every user story of the Product Backlog is traced to at least one functional requirement.

Table C.1: User story to requirement traceability

Story	Epic	Functional requirements	Release
US-001	E01	FR-ACC-01, FR-ACC-07	R1
US-002	E01	FR-ACC-02, FR-ACC-03	R1
US-003	E01	FR-ACC-06	R1
US-010	E02	FR-CNV-01	R1
US-011	E02	FR-CNV-02, FR-CNV-03	R1
US-012	E02	FR-CNV-04	R1
US-013	E02	FR-CNV-05	R1
US-020	E03	FR-WEB-01	R1
US-021	E03	FR-WEB-02	R1
US-022	E03	FR-WEB-03	R1
US-030	E04	FR-KB-01, FR-KB-02	R1
US-031	E04	FR-KB-03, FR-KB-04	R1
US-032	E04	FR-KB-05	R1
US-033	E04	FR-KB-06	R1
US-040	E05	FR-IA-01	R1
US-041	E05	FR-IA-03	R1
US-042	E05	FR-IA-04	R1
US-043	E05	FR-IA-05	R1
US-050	E06	FR-OUT-01	R1
US-051	E06	FR-OUT-02, FR-OUT-03	R1
US-052	E06	FR-OUT-04	R1
US-053	E06	FR-OUT-05	R1
US-060	E07	FR-HUM-01	R1
US-061	E07	FR-HUM-02	R1
US-062	E07	FR-HUM-04	R1
US-063	E07	FR-HUM-05	R1
US-070	E08	FR-TKT-01	R1
US-071	E08	FR-TKT-02	R1
US-080	E09	FR-VOX-01	R2
US-081	E09	FR-VOX-02	R2
US-082	E09	FR-VOX-03	R2
US-083	E09	FR-VOX-04, FR-VOX-05	R2
US-084	E09	FR-VOX-06	R2
US-090	E10	FR-WKF-01	R3
US-091	E10	FR-WKF-02	R3
US-092	E10	FR-WKF-03	R3

Story	Epic	Functional requirements	Release
US-100	E11	FR-INT-01	R3
US-101	E11	FR-INT-02	R3
US-102	E11	FR-INT-03	R3
US-110	E12	FR-OBS-01	R1
US-111	E12	FR-OBS-02	R1
US-112	E12	FR-OBS-03	R1
US-113	E12	FR-OBS-04	R3

Appendix D

Project video

A short motion-graphics video with a voice-over introduces the platform: the customer message, grounded answers (RAG), safe tool calls, prompt-injection handling, tenant isolation, human handoff, voice and the modular monolith. It lasts 1 minute 31 seconds and is published with the presentations.

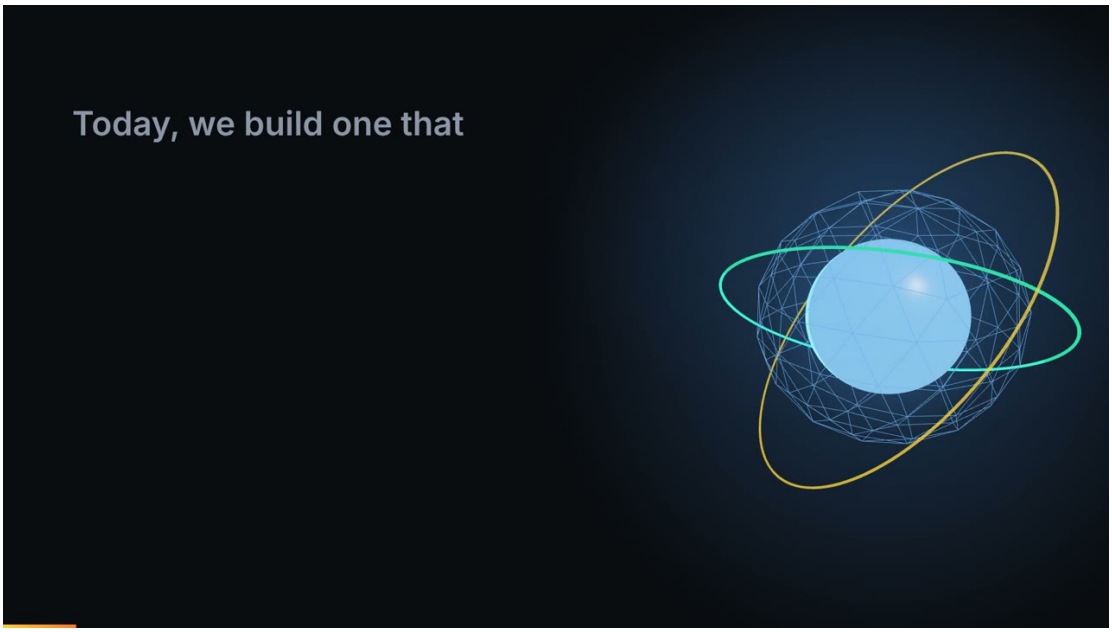


Figure D.1: Poster frame of the project video (click to open the video).

- **Watch online:** `https://cs-agent.messid.tech/assets/ai-customer-support-video.mp4`
- **Presentations page:** `https://cs-agent.messid.tech/` (link “Project video”).
- **File in the repository:** `docs/assets/ai-customer-support-video.mp4` (H.264, 1920×1080, 30 fps).

Appendix E

Open items

Table E.1: Items to be decided or measured

Item	To be decided
Numeric targets	Latency, availability, recovery, volumes, success-measure thresholds (Table 2.1).
Team and calendar	Team size, capacity, Sprint dates (the Gantt dates are placeholders).
Customer verification	Method used before exposing order or account data (ADR-009).
Voice strategy	Modular STT/TTS versus speech-to-speech; provider and telephony (ADR-006, ADR-007).
Retention periods	Messages, transcripts, audio, logs, audit (ADR-008).
Languages and accessibility	Supported customer languages and accessibility level of the widget.
Business tool	Which commerce system backs `orders.lookup` in the MVP.
Names on the cover	Reviewers and approvers of this document.